Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 12:38:47

Measurement Details:

- **Sample:** Muestra12
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

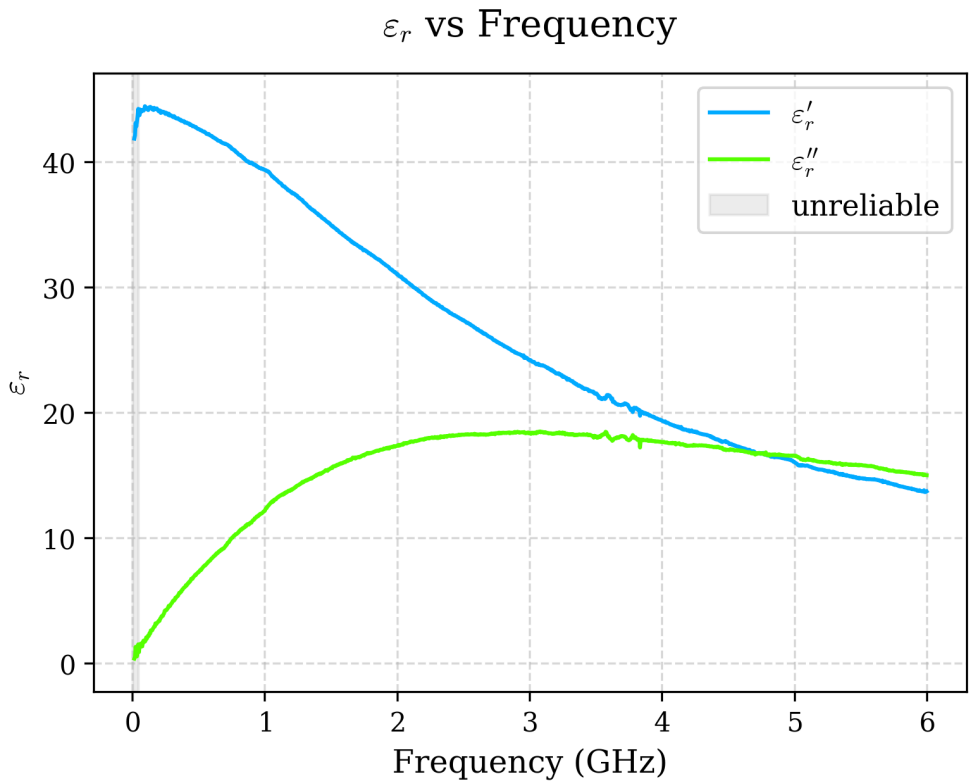
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	41.8717	0.3718	0.0089
0.020	42.1339	0.5053	0.0120
0.025	43.1277	1.3129	0.0304
0.030	42.8365	0.9157	0.0214
0.035	43.1924	0.5517	0.0128
0.040	43.6701	1.2315	0.0282
0.045	44.2458	1.5126	0.0342
0.050	43.7393	0.8847	0.0202
0.055	43.7465	1.3767	0.0315
0.060	44.1945	1.3012	0.0294
0.065	44.0017	1.5773	0.0358
0.070	44.1577	1.4678	0.0332
0.075	44.0886	1.3469	0.0305
0.080	44.0991	1.7033	0.0386
0.085	44.1037	1.8132	0.0411

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	44.2064	1.7127	0.0387
0.095	44.4587	1.7271	0.0388
0.100	44.4089	1.8188	0.0410
0.105	44.2473	2.0654	0.0467
0.110	44.2520	1.9666	0.0444
0.115	44.2922	2.0933	0.0473
0.120	44.1502	2.2218	0.0503
0.125	44.3480	2.1968	0.0495
0.130	44.3485	2.4052	0.0542
0.135	44.4056	2.3741	0.0535
0.140	44.3737	2.6299	0.0593
0.145	44.2932	2.6019	0.0587
0.150	44.2099	2.6290	0.0595
0.155	44.3149	2.7533	0.0621
0.160	44.1554	2.8326	0.0641
0.165	44.3269	2.9314	0.0661
0.170	44.2389	2.9433	0.0665
0.175	44.2125	3.1860	0.0721
0.180	44.2444	3.1147	0.0704
0.185	44.2797	3.1896	0.0720
0.190	44.1655	3.2591	0.0738
0.195	44.1284	3.3182	0.0752
0.200	44.0435	3.3905	0.0770
0.205	44.1256	3.3796	0.0766
0.210	44.0993	3.5541	0.0806
0.215	43.9210	3.6030	0.0820
0.220	44.0472	3.7373	0.0848
0.225	44.0256	3.7866	0.0860
0.230	43.9763	3.7729	0.0858
0.235	44.0123	3.8970	0.0885
0.240	43.9791	3.9944	0.0908
0.245	43.9339	4.1904	0.0954
0.250	43.8438	4.2348	0.0966
0.255	43.9421	4.2241	0.0961
0.260	43.9096	4.2755	0.0974
0.265	43.8451	4.3518	0.0993
0.270	43.7707	4.4452	0.1016
0.275	43.8646	4.5836	0.1045
0.280	43.7201	4.5362	0.1038
0.285	43.7704	4.5824	0.1047
0.290	43.7167	4.6921	0.1073
0.295	43.6973	4.7693	0.1091
0.300	43.6855	4.8277	0.1105
0.305	43.6146	4.9450	0.1134
0.310	43.5830	5.0138	0.1150
0.315	43.6674	5.0041	0.1146

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	43.6111	5.0593	0.1160
0.325	43.5327	5.2267	0.1201
0.330	43.4876	5.2222	0.1201
0.335	43.4769	5.2975	0.1218
0.340	43.4728	5.3607	0.1233
0.345	43.4972	5.3827	0.1237
0.350	43.4252	5.5159	0.1270
0.355	43.3826	5.6268	0.1297
0.360	43.3848	5.6089	0.1293
0.365	43.3793	5.6933	0.1312
0.370	43.2829	5.7220	0.1322
0.375	43.2939	5.8508	0.1351
0.380	43.2779	5.8832	0.1359
0.385	43.2585	5.9757	0.1381
0.390	43.2674	6.0298	0.1394
0.395	43.1575	6.0692	0.1406
0.400	43.1362	6.1186	0.1418
0.405	43.1650	6.2368	0.1445
0.410	43.1615	6.2088	0.1439
0.415	43.1012	6.2899	0.1459
0.420	43.0746	6.3812	0.1481
0.425	43.0897	6.4041	0.1486
0.430	43.0701	6.5101	0.1512
0.435	42.9685	6.5137	0.1516
0.440	42.9990	6.6030	0.1536
0.445	42.9543	6.7459	0.1570
0.450	42.9263	6.7446	0.1571
0.455	42.8588	6.7451	0.1574
0.460	42.8913	6.8594	0.1599
0.465	42.8508	6.9135	0.1613
0.470	42.8312	6.9726	0.1628
0.475	42.7771	7.0591	0.1650
0.480	42.7957	7.0515	0.1648
0.485	42.7219	7.1317	0.1669
0.490	42.6980	7.1874	0.1683
0.495	42.6422	7.2436	0.1699
0.500	42.6380	7.2624	0.1703
0.505	42.6166	7.3451	0.1724
0.510	42.5859	7.3742	0.1732
0.515	42.5418	7.4708	0.1756
0.520	42.5438	7.4524	0.1752
0.525	42.5086	7.5405	0.1774
0.530	42.4656	7.6238	0.1795
0.535	42.4341	7.6788	0.1810
0.540	42.4337	7.7386	0.1824
0.545	42.4226	7.7994	0.1838

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	42.3546	7.8031	0.1842
0.555	42.3322	7.8740	0.1860
0.560	42.2494	7.9160	0.1874
0.565	42.2370	7.9782	0.1889
0.570	42.2160	8.0548	0.1908
0.575	42.2503	8.0938	0.1916
0.580	42.2138	8.1634	0.1934
0.585	42.1446	8.1817	0.1941
0.590	42.1209	8.2366	0.1955
0.595	42.0911	8.3296	0.1979
0.600	42.0849	8.3744	0.1990
0.605	42.0243	8.4141	0.2002
0.610	41.9665	8.4749	0.2019
0.615	41.9599	8.5402	0.2035
0.620	41.9273	8.5634	0.2042
0.625	41.8793	8.6338	0.2062
0.630	41.8699	8.6577	0.2068
0.635	41.8242	8.7666	0.2096
0.640	41.7883	8.7665	0.2098
0.645	41.7206	8.8036	0.2110
0.650	41.7051	8.8849	0.2130
0.655	41.6965	8.9323	0.2142
0.660	41.6728	8.9322	0.2143
0.665	41.6657	8.9968	0.2159
0.670	41.5885	9.0659	0.2180
0.675	41.5700	9.0550	0.2178
0.680	41.5266	9.1287	0.2198
0.685	41.5100	9.1440	0.2203
0.690	41.5059	9.1586	0.2207
0.695	41.4690	9.2243	0.2224
0.700	41.4441	9.2959	0.2243
0.705	41.4594	9.3870	0.2264
0.710	41.4136	9.4140	0.2273
0.715	41.3982	9.5045	0.2296
0.720	41.3749	9.6145	0.2324
0.725	41.3252	9.6452	0.2334
0.730	41.3059	9.6709	0.2341
0.735	41.2197	9.7820	0.2373
0.740	41.2249	9.8441	0.2388
0.745	41.1843	9.9279	0.2411
0.750	41.0621	9.9844	0.2432
0.755	41.0436	10.0142	0.2440
0.760	41.0325	10.0864	0.2458
0.765	40.9835	10.1463	0.2476
0.770	40.9219	10.1976	0.2492
0.775	40.9353	10.2199	0.2497

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	40.8379	10.3035	0.2523
0.785	40.7768	10.3079	0.2528
0.790	40.7756	10.3331	0.2534
0.795	40.7144	10.3871	0.2551
0.800	40.6505	10.4271	0.2565
0.805	40.5833	10.5236	0.2593
0.810	40.5720	10.5484	0.2600
0.815	40.5693	10.6343	0.2621
0.820	40.5021	10.6787	0.2637
0.825	40.4411	10.6640	0.2637
0.830	40.4402	10.7392	0.2656
0.835	40.3674	10.7903	0.2673
0.840	40.2908	10.7941	0.2679
0.845	40.3039	10.8329	0.2688
0.850	40.2089	10.8740	0.2704
0.855	40.1848	10.9835	0.2733
0.860	40.1117	10.9438	0.2728
0.865	40.0583	11.0677	0.2763
0.870	40.0572	11.1064	0.2773
0.875	40.0568	11.0719	0.2764
0.880	39.9700	11.0671	0.2769
0.885	39.9315	11.1299	0.2787
0.890	39.9470	11.2162	0.2808
0.895	39.8987	11.2051	0.2808
0.900	39.9587	11.2255	0.2809
0.905	39.9304	11.3284	0.2837
0.910	39.8785	11.4056	0.2860
0.915	39.8016	11.4377	0.2874
0.920	39.7957	11.4590	0.2879
0.925	39.8024	11.5756	0.2908
0.930	39.7661	11.5988	0.2917
0.935	39.6874	11.6345	0.2932
0.940	39.6922	11.7169	0.2952
0.945	39.6236	11.7288	0.2960
0.950	39.6109	11.7353	0.2963
0.955	39.5686	11.8247	0.2988
0.960	39.5200	11.8433	0.2997
0.965	39.5209	11.9191	0.3016
0.970	39.5222	11.9328	0.3019
0.975	39.4565	11.9426	0.3027
0.980	39.4727	11.9911	0.3038
0.985	39.4688	12.0284	0.3048
0.990	39.4065	12.0927	0.3069
0.995	39.4100	12.1378	0.3080
1.000	39.3852	12.1583	0.3087
1.005	39.3539	12.2754	0.3119

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	39.3434	12.3738	0.3145
1.015	39.2809	12.3935	0.3155
1.020	39.2619	12.5004	0.3184
1.025	39.2700	12.5558	0.3197
1.030	39.1900	12.6108	0.3218
1.035	39.1782	12.6752	0.3235
1.040	39.0984	12.7421	0.3259
1.045	39.0181	12.7955	0.3279
1.050	38.9612	12.8672	0.3303
1.055	38.8618	12.8550	0.3308
1.060	38.8575	12.9582	0.3335
1.065	38.7895	12.9112	0.3329
1.070	38.7711	13.0212	0.3358
1.075	38.6830	13.0735	0.3380
1.080	38.6620	13.0663	0.3380
1.085	38.6281	13.0864	0.3388
1.090	38.5887	13.1720	0.3413
1.095	38.5051	13.2241	0.3434
1.100	38.4818	13.2671	0.3448
1.105	38.4240	13.2762	0.3455
1.110	38.3621	13.3387	0.3477
1.115	38.3342	13.3800	0.3490
1.120	38.3194	13.3922	0.3495
1.125	38.2689	13.4132	0.3505
1.130	38.1918	13.4721	0.3527
1.135	38.1047	13.4897	0.3540
1.140	38.0820	13.4604	0.3535
1.145	38.0449	13.5585	0.3564
1.150	38.0531	13.5321	0.3556
1.155	37.9831	13.6234	0.3587
1.160	37.9169	13.5978	0.3586
1.165	37.8914	13.6763	0.3609
1.170	37.8481	13.6791	0.3614
1.175	37.7577	13.7035	0.3629
1.180	37.7428	13.7726	0.3649
1.185	37.7334	13.7608	0.3647
1.190	37.6706	13.7600	0.3653
1.195	37.6436	13.8154	0.3670
1.200	37.6260	13.8657	0.3685
1.205	37.5252	13.8355	0.3687
1.210	37.5517	13.8974	0.3701
1.215	37.4809	13.9369	0.3718
1.220	37.4943	13.9537	0.3722
1.225	37.4627	13.9957	0.3736
1.230	37.4241	14.0627	0.3758
1.235	37.3388	14.0705	0.3768

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	37.3471	14.1386	0.3786
1.245	37.3196	14.1752	0.3798
1.250	37.2159	14.1972	0.3815
1.255	37.1631	14.2201	0.3826
1.260	37.1604	14.2480	0.3834
1.265	37.0848	14.3005	0.3856
1.270	37.0590	14.3159	0.3863
1.275	37.0236	14.3681	0.3881
1.280	36.9710	14.4102	0.3898
1.285	36.9633	14.4746	0.3916
1.290	36.8631	14.4309	0.3915
1.295	36.7999	14.4914	0.3938
1.300	36.7651	14.5228	0.3950
1.305	36.7201	14.5404	0.3960
1.310	36.6762	14.5882	0.3978
1.315	36.6362	14.6106	0.3988
1.320	36.5569	14.5929	0.3992
1.325	36.5020	14.6530	0.4014
1.330	36.4433	14.6667	0.4025
1.335	36.4071	14.6849	0.4034
1.340	36.3937	14.7175	0.4044
1.345	36.2953	14.7268	0.4057
1.350	36.2659	14.7803	0.4076
1.355	36.2158	14.7406	0.4070
1.360	36.2135	14.8112	0.4090
1.365	36.1526	14.8523	0.4108
1.370	36.1162	14.8614	0.4115
1.375	36.1163	14.9130	0.4129
1.380	36.0253	14.9349	0.4146
1.385	35.9834	14.9912	0.4166
1.390	35.9377	14.9745	0.4167
1.395	35.8978	15.0564	0.4194
1.400	35.8920	15.0629	0.4197
1.405	35.7741	15.0891	0.4218
1.410	35.7896	15.1281	0.4227
1.415	35.7178	15.0843	0.4223
1.420	35.7004	15.1570	0.4246
1.425	35.6639	15.1200	0.4240
1.430	35.5679	15.1877	0.4270
1.435	35.5461	15.1909	0.4274
1.440	35.4961	15.2401	0.4293
1.445	35.4622	15.2663	0.4305
1.450	35.4574	15.3063	0.4317
1.455	35.3838	15.3533	0.4339
1.460	35.3469	15.4095	0.4360
1.465	35.2886	15.4120	0.4367

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	35.2380	15.4063	0.4372
1.475	35.2119	15.3607	0.4362
1.480	35.1941	15.4602	0.4393
1.485	35.1220	15.4737	0.4406
1.490	35.1121	15.5028	0.4415
1.495	35.0377	15.5293	0.4432
1.500	34.9752	15.5861	0.4456
1.505	34.9302	15.6143	0.4470
1.510	34.8698	15.5842	0.4469
1.515	34.8862	15.7035	0.4501
1.520	34.7998	15.7004	0.4512
1.525	34.7595	15.7429	0.4529
1.530	34.7340	15.7191	0.4526
1.535	34.6987	15.7443	0.4537
1.540	34.6322	15.7592	0.4550
1.545	34.5728	15.8373	0.4581
1.550	34.5808	15.8354	0.4579
1.555	34.5247	15.8335	0.4586
1.560	34.4568	15.8783	0.4608
1.565	34.3965	15.9171	0.4628
1.570	34.3618	15.9110	0.4630
1.575	34.3308	15.9663	0.4651
1.580	34.3183	15.9491	0.4647
1.585	34.2467	15.9550	0.4659
1.590	34.1908	16.0068	0.4682
1.595	34.1660	16.0156	0.4688
1.600	34.1201	16.0547	0.4705
1.605	34.0978	16.0650	0.4711
1.610	34.0498	16.0501	0.4714
1.615	33.9965	16.1165	0.4741
1.620	33.9643	16.1679	0.4760
1.625	33.9256	16.2109	0.4778
1.630	33.8679	16.1492	0.4768
1.635	33.8397	16.2137	0.4791
1.640	33.8183	16.2371	0.4801
1.645	33.7648	16.2568	0.4815
1.650	33.6927	16.2340	0.4818
1.655	33.6798	16.2636	0.4829
1.660	33.5905	16.3239	0.4860
1.665	33.6164	16.4173	0.4884
1.670	33.5454	16.3956	0.4888
1.675	33.4881	16.3717	0.4889
1.680	33.5087	16.4106	0.4897
1.685	33.4560	16.3408	0.4884
1.690	33.4188	16.4240	0.4915
1.695	33.4078	16.4124	0.4913

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	33.3483	16.4390	0.4929
1.705	33.2995	16.4643	0.4944
1.710	33.2758	16.4602	0.4947
1.715	33.2635	16.4317	0.4940
1.720	33.1755	16.4943	0.4972
1.725	33.1471	16.4799	0.4972
1.730	33.1789	16.5420	0.4986
1.735	33.1009	16.5538	0.5001
1.740	33.0418	16.5963	0.5023
1.745	32.9565	16.6177	0.5042
1.750	33.0118	16.5731	0.5020
1.755	32.9776	16.6653	0.5054
1.760	32.9436	16.6600	0.5057
1.765	32.9218	16.7023	0.5073
1.770	32.8564	16.7104	0.5086
1.775	32.8397	16.7857	0.5111
1.780	32.7991	16.7368	0.5103
1.785	32.7212	16.7857	0.5130
1.790	32.7122	16.7552	0.5122
1.795	32.7018	16.7883	0.5134
1.800	32.6626	16.7708	0.5135
1.805	32.5870	16.7967	0.5154
1.810	32.5787	16.8087	0.5159
1.815	32.5431	16.8261	0.5170
1.820	32.5113	16.8483	0.5182
1.825	32.4848	16.8748	0.5195
1.830	32.4064	16.8841	0.5210
1.835	32.3795	16.9239	0.5227
1.840	32.3360	16.9377	0.5238
1.845	32.3048	16.9593	0.5250
1.850	32.2732	16.9678	0.5258
1.855	32.2552	16.9744	0.5263
1.860	32.2040	16.9511	0.5264
1.865	32.2035	17.0026	0.5280
1.870	32.1698	17.0195	0.5291
1.875	32.1051	17.0494	0.5310
1.880	32.0766	17.0507	0.5316
1.885	32.0025	17.0444	0.5326
1.890	31.9786	17.0548	0.5333
1.895	31.9349	17.0719	0.5346
1.900	31.8797	17.1414	0.5377
1.905	31.8678	17.1181	0.5372
1.910	31.8226	17.1231	0.5381
1.915	31.7601	17.1639	0.5404
1.920	31.7342	17.1484	0.5404
1.925	31.7096	17.2025	0.5425

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	31.6705	17.1476	0.5414
1.935	31.6349	17.2060	0.5439
1.940	31.5784	17.1379	0.5427
1.945	31.4951	17.2312	0.5471
1.950	31.4621	17.2314	0.5477
1.955	31.4222	17.2645	0.5494
1.960	31.4234	17.2403	0.5486
1.965	31.3323	17.2580	0.5508
1.970	31.3094	17.2583	0.5512
1.975	31.2639	17.3319	0.5544
1.980	31.2087	17.2845	0.5538
1.985	31.2156	17.2890	0.5539
1.990	31.1584	17.3235	0.5560
1.995	31.1081	17.3449	0.5576
2.000	31.0256	17.3672	0.5598
2.005	31.0471	17.3585	0.5591
2.010	30.9697	17.4293	0.5628
2.015	30.9159	17.3649	0.5617
2.020	30.8732	17.3920	0.5633
2.025	30.8553	17.4352	0.5651
2.030	30.8636	17.4221	0.5645
2.035	30.7792	17.4751	0.5678
2.040	30.7568	17.4778	0.5683
2.045	30.6443	17.4464	0.5693
2.050	30.6543	17.5681	0.5731
2.055	30.6350	17.5574	0.5731
2.060	30.5558	17.5572	0.5746
2.065	30.5674	17.5839	0.5753
2.070	30.4771	17.5480	0.5758
2.075	30.4172	17.5530	0.5771
2.080	30.3971	17.5915	0.5787
2.085	30.3670	17.6113	0.5799
2.090	30.3304	17.6229	0.5810
2.095	30.2848	17.6357	0.5823
2.100	30.2093	17.6965	0.5858
2.105	30.1889	17.6737	0.5854
2.110	30.1792	17.7101	0.5868
2.115	30.1078	17.6994	0.5879
2.120	30.0491	17.6677	0.5880
2.125	29.9957	17.7074	0.5903
2.130	29.9810	17.7123	0.5908
2.135	29.9503	17.7402	0.5923
2.140	29.9280	17.7480	0.5930
2.145	29.8568	17.7310	0.5939
2.150	29.8676	17.7528	0.5944
2.155	29.7925	17.7643	0.5963

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	29.7246	17.7773	0.5981
2.165	29.7336	17.8126	0.5991
2.170	29.6565	17.8102	0.6005
2.175	29.6374	17.8466	0.6022
2.180	29.5917	17.8454	0.6031
2.185	29.5618	17.8622	0.6042
2.190	29.5061	17.8673	0.6055
2.195	29.4082	17.8554	0.6072
2.200	29.3886	17.8785	0.6083
2.205	29.3779	17.8561	0.6078
2.210	29.3445	17.9323	0.6111
2.215	29.2965	17.9028	0.6111
2.220	29.2590	17.9247	0.6126
2.225	29.2035	17.9614	0.6150
2.230	29.1551	17.9662	0.6162
2.235	29.1398	17.9711	0.6167
2.240	29.1137	17.9444	0.6164
2.245	29.0356	17.9883	0.6195
2.250	29.0184	17.9806	0.6196
2.255	29.0057	17.9891	0.6202
2.260	28.9187	18.0187	0.6231
2.265	28.9009	18.0341	0.6240
2.270	28.8781	17.9799	0.6226
2.275	28.8020	17.9908	0.6246
2.280	28.7925	17.9886	0.6248
2.285	28.7578	18.0561	0.6279
2.290	28.7233	18.0259	0.6276
2.295	28.7373	18.0144	0.6269
2.300	28.6798	17.9786	0.6269
2.305	28.5933	18.0553	0.6315
2.310	28.6099	17.9736	0.6282
2.315	28.5972	18.0474	0.6311
2.320	28.5350	18.0604	0.6329
2.325	28.4687	18.0012	0.6323
2.330	28.4651	18.1194	0.6366
2.335	28.4085	18.0825	0.6365
2.340	28.4350	18.0494	0.6348
2.345	28.3216	18.0900	0.6387
2.350	28.2986	18.1113	0.6400
2.355	28.2782	18.0463	0.6382
2.360	28.2500	18.1207	0.6414
2.365	28.2203	18.1297	0.6424
2.370	28.1957	18.1098	0.6423
2.375	28.1721	18.0766	0.6416
2.380	28.1338	18.1434	0.6449
2.385	28.0764	18.1376	0.6460

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	28.0771	18.1558	0.6466
2.395	27.9896	18.1870	0.6498
2.400	27.9436	18.1708	0.6503
2.405	27.9385	18.1243	0.6487
2.410	27.9275	18.1611	0.6503
2.415	27.8653	18.1586	0.6517
2.420	27.8370	18.2043	0.6540
2.425	27.8044	18.1515	0.6528
2.430	27.7787	18.1530	0.6535
2.435	27.7830	18.1708	0.6540
2.440	27.7310	18.1724	0.6553
2.445	27.7037	18.1918	0.6567
2.450	27.6893	18.1977	0.6572
2.455	27.6393	18.1819	0.6578
2.460	27.6126	18.2002	0.6591
2.465	27.5959	18.2454	0.6612
2.470	27.5395	18.1989	0.6608
2.475	27.5190	18.2092	0.6617
2.480	27.4752	18.2198	0.6631
2.485	27.4690	18.2515	0.6644
2.490	27.4339	18.2331	0.6646
2.495	27.3770	18.2495	0.6666
2.500	27.3921	18.2482	0.6662
2.505	27.3472	18.2329	0.6667
2.510	27.3182	18.2555	0.6683
2.515	27.2738	18.2509	0.6692
2.520	27.2452	18.2752	0.6708
2.525	27.1739	18.2573	0.6719
2.530	27.1648	18.2660	0.6724
2.535	27.1771	18.2834	0.6728
2.540	27.1261	18.2586	0.6731
2.545	27.0975	18.2496	0.6735
2.550	27.0430	18.2626	0.6753
2.555	27.0432	18.2660	0.6754
2.560	26.9815	18.2840	0.6777
2.565	26.9475	18.2732	0.6781
2.570	26.9234	18.2683	0.6785
2.575	26.8979	18.2755	0.6794
2.580	26.8384	18.2576	0.6803
2.585	26.7904	18.3119	0.6835
2.590	26.7827	18.2483	0.6813
2.595	26.7362	18.2781	0.6836
2.600	26.6871	18.2639	0.6844
2.605	26.6244	18.2395	0.6851
2.610	26.6614	18.2643	0.6850
2.615	26.6167	18.2303	0.6849

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	26.5528	18.2602	0.6877
2.625	26.5125	18.2524	0.6884
2.630	26.4990	18.2818	0.6899
2.635	26.4946	18.2560	0.6890
2.640	26.4014	18.2644	0.6918
2.645	26.4025	18.2572	0.6915
2.650	26.3290	18.2693	0.6939
2.655	26.3396	18.2479	0.6928
2.660	26.2729	18.2646	0.6952
2.665	26.3006	18.2678	0.6946
2.670	26.2629	18.2297	0.6941
2.675	26.2111	18.2494	0.6962
2.680	26.1798	18.2610	0.6975
2.685	26.1184	18.2547	0.6989
2.690	26.0938	18.2544	0.6996
2.695	26.0632	18.2780	0.7013
2.700	26.0379	18.2922	0.7025
2.705	26.0176	18.2849	0.7028
2.710	25.9694	18.3323	0.7059
2.715	25.9682	18.2958	0.7045
2.720	25.9032	18.3267	0.7075
2.725	25.8366	18.3008	0.7083
2.730	25.7853	18.3070	0.7100
2.735	25.7680	18.3512	0.7122
2.740	25.7737	18.2784	0.7092
2.745	25.7201	18.3654	0.7140
2.750	25.6715	18.3652	0.7154
2.755	25.6296	18.3590	0.7163
2.760	25.6425	18.3734	0.7165
2.765	25.5841	18.3373	0.7167
2.770	25.5769	18.4108	0.7198
2.775	25.5149	18.3435	0.7189
2.780	25.5048	18.3553	0.7197
2.785	25.4686	18.3824	0.7218
2.790	25.4660	18.3625	0.7211
2.795	25.4140	18.4044	0.7242
2.800	25.3844	18.3902	0.7245
2.805	25.3709	18.4065	0.7255
2.810	25.2916	18.3767	0.7266
2.815	25.2842	18.3980	0.7277
2.820	25.2736	18.3938	0.7278
2.825	25.2282	18.3757	0.7284
2.830	25.2096	18.4048	0.7301
2.835	25.1960	18.3999	0.7303
2.840	25.1332	18.4164	0.7327
2.845	25.1023	18.4243	0.7340

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	25.1025	18.3559	0.7312
2.855	25.0615	18.4141	0.7348
2.860	25.0107	18.3825	0.7350
2.865	24.9561	18.4062	0.7375
2.870	24.9335	18.3978	0.7379
2.875	24.9178	18.3859	0.7379
2.880	24.8825	18.4262	0.7405
2.885	24.8795	18.4250	0.7406
2.890	24.7936	18.4640	0.7447
2.895	24.7917	18.4035	0.7423
2.900	24.7840	18.4099	0.7428
2.905	24.7349	18.4580	0.7462
2.910	24.7237	18.4740	0.7472
2.915	24.6949	18.4402	0.7467
2.920	24.6788	18.4480	0.7475
2.925	24.6738	18.4030	0.7459
2.930	24.6408	18.4143	0.7473
2.935	24.5961	18.4232	0.7490
2.940	24.5981	18.4345	0.7494
2.945	24.5449	18.4375	0.7512
2.950	24.4982	18.4217	0.7520
2.955	24.5764	18.4172	0.7494
2.960	24.3717	18.4555	0.7573
2.965	24.4569	18.3927	0.7520
2.970	24.3802	18.3553	0.7529
2.975	24.3640	18.3713	0.7540
2.980	24.3747	18.4056	0.7551
2.985	24.2978	18.3839	0.7566
2.990	24.2532	18.4257	0.7597
2.995	24.2539	18.3532	0.7567
3.000	24.1697	18.3926	0.7610
3.005	24.2063	18.3837	0.7595
3.010	24.1700	18.3106	0.7576
3.015	24.1312	18.3308	0.7596
3.020	24.0669	18.4097	0.7649
3.025	24.0455	18.3585	0.7635
3.030	24.0603	18.3409	0.7623
3.035	24.0452	18.4092	0.7656
3.040	23.9723	18.3975	0.7675
3.045	23.9277	18.3822	0.7682
3.050	23.8690	18.3459	0.7686
3.055	23.8901	18.4598	0.7727
3.060	23.8572	18.4584	0.7737
3.065	23.9131	18.4140	0.7700
3.070	23.8112	18.4804	0.7761
3.075	23.8342	18.4772	0.7752

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	23.8033	18.4988	0.7772
3.085	23.7748	18.4278	0.7751
3.090	23.7773	18.4605	0.7764
3.095	23.7752	18.4673	0.7767
3.100	23.7835	18.4611	0.7762
3.105	23.7449	18.4046	0.7751
3.110	23.7302	18.4429	0.7772
3.115	23.7111	18.4219	0.7769
3.120	23.6754	18.4451	0.7791
3.125	23.6450	18.3985	0.7781
3.130	23.6126	18.3972	0.7791
3.135	23.5604	18.3868	0.7804
3.140	23.5772	18.3777	0.7795
3.145	23.5098	18.3399	0.7801
3.150	23.5009	18.3568	0.7811
3.155	23.4511	18.3566	0.7828
3.160	23.4568	18.3408	0.7819
3.165	23.3667	18.3526	0.7854
3.170	23.4113	18.3169	0.7824
3.175	23.3392	18.3073	0.7844
3.180	23.3212	18.3202	0.7856
3.185	23.2549	18.3480	0.7890
3.190	23.2511	18.3230	0.7880
3.195	23.1922	18.3572	0.7915
3.200	23.1276	18.3405	0.7930
3.205	23.1382	18.3900	0.7948
3.210	23.1158	18.3732	0.7948
3.215	23.1257	18.3522	0.7936
3.220	23.1461	18.4027	0.7951
3.225	23.0539	18.4298	0.7994
3.230	23.0808	18.3383	0.7945
3.235	23.0353	18.3966	0.7986
3.240	22.9646	18.3733	0.8001
3.245	22.9944	18.3251	0.7969
3.250	22.8687	18.4193	0.8054
3.255	22.8440	18.3301	0.8024
3.260	22.8928	18.3463	0.8014
3.265	22.8512	18.3938	0.8049
3.270	22.8119	18.2640	0.8006
3.275	22.7668	18.2891	0.8033
3.280	22.7720	18.3081	0.8040
3.285	22.6261	18.3163	0.8095
3.290	22.7085	18.2630	0.8042
3.295	22.6292	18.2558	0.8067
3.300	22.6717	18.2648	0.8056
3.305	22.6052	18.2920	0.8092

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	22.5709	18.2590	0.8090
3.315	22.5281	18.2458	0.8099
3.320	22.5198	18.2915	0.8122
3.325	22.5204	18.2574	0.8107
3.330	22.4516	18.2813	0.8143
3.335	22.3723	18.2075	0.8138
3.340	22.3901	18.2437	0.8148
3.345	22.3329	18.2028	0.8151
3.350	22.2710	18.2194	0.8181
3.355	22.2981	18.2114	0.8167
3.360	22.2159	18.2459	0.8213
3.365	22.1875	18.2281	0.8215
3.370	22.1393	18.2329	0.8236
3.375	22.1824	18.2683	0.8235
3.380	22.1449	18.2553	0.8244
3.385	22.0940	18.2438	0.8257
3.390	22.0648	18.2361	0.8265
3.395	22.0669	18.2475	0.8269
3.400	21.9629	18.2633	0.8316
3.405	21.9611	18.2324	0.8302
3.410	21.9584	18.2771	0.8324
3.415	21.8840	18.2491	0.8339
3.420	21.9018	18.2295	0.8323
3.425	21.9129	18.2396	0.8324
3.430	21.8946	18.2406	0.8331
3.435	21.8652	18.2866	0.8363
3.440	21.8609	18.2131	0.8331
3.445	21.7979	18.2419	0.8369
3.450	21.7185	18.2034	0.8382
3.455	21.7660	18.2449	0.8382
3.460	21.7230	18.2259	0.8390
3.465	21.7032	18.1795	0.8376
3.470	21.6889	18.1702	0.8378
3.475	21.6929	18.1342	0.8360
3.480	21.6452	18.0820	0.8354
3.485	21.6576	18.0045	0.8313
3.490	21.5429	18.1572	0.8428
3.495	21.5196	18.1372	0.8428
3.500	21.5076	18.0603	0.8397
3.505	21.5406	18.1071	0.8406
3.510	21.4807	17.9965	0.8378
3.515	21.4031	18.0752	0.8445
3.520	21.3510	18.0208	0.8440
3.525	21.1703	18.0851	0.8543
3.530	21.2317	18.1317	0.8540
3.535	21.1925	18.0604	0.8522

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	21.0981	18.1312	0.8594
3.545	21.1548	18.2349	0.8620
3.550	21.0778	18.2396	0.8653
3.555	21.1221	18.2278	0.8630
3.560	21.1413	18.3024	0.8657
3.565	21.1974	18.2845	0.8626
3.570	21.0927	18.3348	0.8692
3.575	21.1276	18.4724	0.8743
3.580	21.2390	18.4437	0.8684
3.585	21.4054	18.3328	0.8565
3.590	21.4074	18.1559	0.8481
3.595	21.4147	18.0766	0.8441
3.600	21.3335	18.1182	0.8493
3.605	21.3615	17.9678	0.8411
3.610	21.2905	17.9205	0.8417
3.615	21.1618	17.9078	0.8462
3.620	21.0601	17.8613	0.8481
3.625	21.1406	17.7544	0.8398
3.630	20.8542	17.7904	0.8531
3.635	20.8547	17.8760	0.8572
3.640	20.8739	17.9267	0.8588
3.645	20.8131	17.9537	0.8626
3.650	20.7255	17.9731	0.8672
3.655	20.7079	17.9507	0.8669
3.660	20.6812	18.0352	0.8721
3.665	20.6711	17.9907	0.8703
3.670	20.6178	18.0439	0.8752
3.675	20.6299	18.0041	0.8727
3.680	20.6286	18.0496	0.8750
3.685	20.6176	18.1062	0.8782
3.690	20.5975	18.1003	0.8788
3.695	20.6119	18.1341	0.8798
3.700	20.6389	18.1334	0.8786
3.705	20.6689	18.1228	0.8768
3.710	20.6343	18.1108	0.8777
3.715	20.6905	18.1029	0.8749
3.720	20.7502	18.0466	0.8697
3.725	20.7032	18.0536	0.8720
3.730	20.7105	17.9742	0.8679
3.735	20.6597	17.8796	0.8654
3.740	20.6530	17.8477	0.8642
3.745	20.5942	17.8237	0.8655
3.750	20.4816	17.7909	0.8686
3.755	20.3994	17.8114	0.8731
3.760	20.3062	17.8539	0.8792
3.765	20.2100	17.9262	0.8870

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	20.2573	18.0328	0.8902
3.775	20.3145	17.9972	0.8859
3.780	20.0478	18.2138	0.9085
3.785	20.3135	18.1176	0.8919
3.790	20.3959	17.9839	0.8817
3.795	20.3735	17.9702	0.8820
3.800	20.3343	18.0492	0.8876
3.805	20.3862	17.9477	0.8804
3.810	20.2922	17.9253	0.8834
3.815	20.2509	17.8238	0.8801
3.820	20.2766	17.8974	0.8827
3.825	20.2110	17.8219	0.8818
3.830	20.2054	17.8570	0.8838
3.835	19.7549	17.2208	0.8717
3.840	20.1156	17.8316	0.8865
3.845	20.0843	17.8438	0.8884
3.850	20.0702	17.8209	0.8879
3.855	20.0226	17.8438	0.8912
3.860	19.9958	17.7933	0.8899
3.865	19.9168	17.8093	0.8942
3.870	19.9830	17.8094	0.8912
3.875	19.9159	17.7984	0.8937
3.880	19.9406	17.7924	0.8923
3.885	19.8673	17.7974	0.8958
3.890	19.8803	17.8320	0.8970
3.895	19.8816	17.7557	0.8931
3.900	19.8268	17.7523	0.8954
3.905	19.7638	17.7956	0.9004
3.910	19.7951	17.7643	0.8974
3.915	19.7595	17.7546	0.8985
3.920	19.7237	17.7898	0.9019
3.925	19.6985	17.7376	0.9005
3.930	19.6849	17.7379	0.9011
3.935	19.6894	17.7363	0.9008
3.940	19.6362	17.7334	0.9031
3.945	19.6258	17.7122	0.9025
3.950	19.5812	17.6960	0.9037
3.955	19.5473	17.7157	0.9063
3.960	19.5431	17.6925	0.9053
3.965	19.5330	17.6970	0.9060
3.970	19.4890	17.7085	0.9086
3.975	19.4654	17.6897	0.9088
3.980	19.4514	17.6757	0.9087
3.985	19.4451	17.6904	0.9098
3.990	19.4370	17.6775	0.9095
3.995	19.3964	17.6545	0.9102

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	19.3674	17.6606	0.9119
4.005	19.3400	17.6517	0.9127
4.010	19.2989	17.6473	0.9144
4.015	19.3020	17.6709	0.9155
4.020	19.2734	17.6337	0.9149
4.025	19.3076	17.6129	0.9122
4.030	19.2107	17.6150	0.9169
4.035	19.1890	17.6309	0.9188
4.040	19.2114	17.6219	0.9173
4.045	19.1955	17.5871	0.9162
4.050	19.1620	17.5737	0.9171
4.055	19.1232	17.5626	0.9184
4.060	19.0864	17.5686	0.9205
4.065	19.0565	17.5689	0.9219
4.070	19.0612	17.5871	0.9227
4.075	19.0325	17.5599	0.9226
4.080	19.0041	17.5999	0.9261
4.085	18.9891	17.5655	0.9250
4.090	18.9938	17.5553	0.9243
4.095	18.9358	17.5718	0.9280
4.100	18.9424	17.5921	0.9287
4.105	18.9079	17.4609	0.9235
4.110	18.9117	17.5498	0.9280
4.115	18.8728	17.5367	0.9292
4.120	18.8370	17.5328	0.9308
4.125	18.8862	17.5256	0.9280
4.130	18.8584	17.5668	0.9315
4.135	18.7795	17.5059	0.9322
4.140	18.7974	17.4549	0.9286
4.145	18.8655	17.4552	0.9252
4.150	18.7101	17.4838	0.9345
4.155	18.6766	17.4898	0.9365
4.160	18.7157	17.4865	0.9343
4.165	18.6557	17.4298	0.9343
4.170	18.6347	17.4228	0.9350
4.175	18.6582	17.3767	0.9313
4.180	18.6557	17.4193	0.9337
4.185	18.6320	17.3521	0.9313
4.190	18.5851	17.3531	0.9337
4.195	18.5770	17.3512	0.9340
4.200	18.6043	17.3634	0.9333
4.205	18.5937	17.3633	0.9338
4.210	18.5024	17.3600	0.9383
4.215	18.5531	17.3740	0.9364
4.220	18.5274	17.3743	0.9378
4.225	18.5889	17.3816	0.9351

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	18.5246	17.3665	0.9375
4.235	18.4816	17.3607	0.9393
4.240	18.4699	17.4336	0.9439
4.245	18.5037	17.3856	0.9396
4.250	18.4701	17.3646	0.9401
4.255	18.4761	17.4061	0.9421
4.260	18.4166	17.3861	0.9440
4.265	18.4703	17.3811	0.9410
4.270	18.4002	17.3778	0.9444
4.275	18.3789	17.3746	0.9454
4.280	18.3676	17.3921	0.9469
4.285	18.3428	17.3973	0.9485
4.290	18.3253	17.3851	0.9487
4.295	18.3157	17.3778	0.9488
4.300	18.3041	17.4009	0.9507
4.305	18.2797	17.3878	0.9512
4.310	18.2775	17.3760	0.9507
4.315	18.2610	17.3816	0.9518
4.320	18.2616	17.3696	0.9512
4.325	18.2430	17.3735	0.9523
4.330	18.2446	17.3710	0.9521
4.335	18.2256	17.3713	0.9531
4.340	18.2273	17.3410	0.9514
4.345	18.1900	17.3349	0.9530
4.350	18.1883	17.3348	0.9531
4.355	18.1631	17.3411	0.9547
4.360	18.1375	17.3075	0.9542
4.365	18.0905	17.3026	0.9564
4.370	18.1210	17.3011	0.9548
4.375	18.0451	17.3113	0.9593
4.380	18.0557	17.2950	0.9579
4.385	17.9778	17.2783	0.9611
4.390	18.0077	17.2477	0.9578
4.395	17.9513	17.2547	0.9612
4.400	17.9343	17.2733	0.9631
4.405	17.8767	17.2597	0.9655
4.410	17.8965	17.2724	0.9651
4.415	17.8711	17.2336	0.9643
4.420	17.8403	17.2830	0.9688
4.425	17.8633	17.2897	0.9679
4.430	17.8146	17.2371	0.9676
4.435	17.7675	17.2005	0.9681
4.440	17.7817	17.2294	0.9689
4.445	17.7077	17.1916	0.9709
4.450	17.7315	17.1563	0.9676
4.455	17.6556	17.1987	0.9741

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	17.6845	17.1109	0.9676
4.465	17.6907	17.0562	0.9641
4.470	17.7049	17.0877	0.9651
4.475	17.7191	17.0484	0.9621
4.480	17.7426	17.0691	0.9620
4.485	17.6449	17.0082	0.9639
4.490	17.6595	17.0426	0.9651
4.495	17.6583	17.0364	0.9648
4.500	17.6188	17.0363	0.9669
4.505	17.5734	17.0192	0.9685
4.510	17.4961	17.0163	0.9726
4.515	17.5058	17.0825	0.9758
4.520	17.4874	17.0700	0.9761
4.525	17.5111	17.0542	0.9739
4.530	17.4390	17.0276	0.9764
4.535	17.4380	17.0542	0.9780
4.540	17.4361	17.0210	0.9762
4.545	17.3882	17.0147	0.9785
4.550	17.3894	16.9725	0.9760
4.555	17.3448	16.9974	0.9800
4.560	17.3488	17.0029	0.9801
4.565	17.3196	17.0169	0.9825
4.570	17.3041	16.9784	0.9812
4.575	17.3029	16.9770	0.9812
4.580	17.2887	16.9966	0.9831
4.585	17.2534	16.9788	0.9841
4.590	17.2531	16.9556	0.9828
4.595	17.2249	16.9725	0.9853
4.600	17.2097	16.9483	0.9848
4.605	17.1811	16.9368	0.9858
4.610	17.1735	16.9493	0.9869
4.615	17.1566	16.9240	0.9864
4.620	17.1510	16.9035	0.9856
4.625	17.1262	16.9172	0.9878
4.630	17.1196	16.8923	0.9867
4.635	17.0896	16.8684	0.9871
4.640	17.0552	16.8716	0.9892
4.645	17.0517	16.8570	0.9886
4.650	17.0432	16.8489	0.9886
4.655	17.0226	16.8149	0.9878
4.660	17.0438	16.8518	0.9887
4.665	17.0114	16.8401	0.9899
4.670	16.9242	16.8153	0.9936
4.675	16.9323	16.7543	0.9895
4.680	16.9239	16.7666	0.9907
4.685	16.9236	16.7501	0.9898

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	16.8995	16.8112	0.9948
4.695	16.9148	16.7373	0.9895
4.700	16.8599	16.7782	0.9952
4.705	16.8183	16.6544	0.9903
4.710	16.7383	16.7544	1.0010
4.715	16.7785	16.7594	0.9989
4.720	16.7647	16.7087	0.9967
4.725	16.7649	16.7745	1.0006
4.730	16.7848	16.7355	0.9971
4.735	16.7868	16.7696	0.9990
4.740	16.7850	16.7487	0.9978
4.745	16.7133	16.7553	1.0025
4.750	16.7761	16.7249	0.9970
4.755	16.7559	16.7695	1.0008
4.760	16.7212	16.7148	0.9996
4.765	16.7893	16.7417	0.9972
4.770	16.7536	16.7638	1.0006
4.775	16.6523	16.6607	1.0005
4.780	16.6818	16.6707	0.9993
4.785	16.6645	16.8328	1.0101
4.790	16.6702	16.6379	0.9981
4.795	16.5953	16.6445	1.0030
4.800	16.6048	16.6503	1.0027
4.805	16.6308	16.5869	0.9974
4.810	16.5747	16.6775	1.0062
4.815	16.6049	16.5813	0.9986
4.820	16.4972	16.6456	1.0090
4.825	16.5420	16.6495	1.0065
4.830	16.4776	16.6477	1.0103
4.835	16.4808	16.6644	1.0111
4.840	16.4918	16.6564	1.0100
4.845	16.5114	16.6315	1.0073
4.850	16.4367	16.6497	1.0130
4.855	16.4906	16.6760	1.0112
4.860	16.4662	16.6392	1.0105
4.865	16.5061	16.6327	1.0077
4.870	16.4451	16.6245	1.0109
4.875	16.4547	16.6432	1.0115
4.880	16.4438	16.6632	1.0133
4.885	16.4305	16.6120	1.0110
4.890	16.4805	16.6603	1.0109
4.895	16.4089	16.6130	1.0124
4.900	16.4176	16.6386	1.0135
4.905	16.4131	16.6170	1.0124
4.910	16.3703	16.6067	1.0144
4.915	16.3693	16.6113	1.0148

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	16.3636	16.5989	1.0144
4.925	16.3562	16.5941	1.0145
4.930	16.3192	16.5793	1.0159
4.935	16.3221	16.5810	1.0159
4.940	16.3169	16.5758	1.0159
4.945	16.2874	16.5753	1.0177
4.950	16.2859	16.5734	1.0177
4.955	16.2775	16.5844	1.0189
4.960	16.2571	16.5614	1.0187
4.965	16.2499	16.5657	1.0194
4.970	16.2287	16.5754	1.0214
4.975	16.2362	16.5761	1.0209
4.980	16.1758	16.5750	1.0247
4.985	16.1675	16.5553	1.0240
4.990	16.1436	16.5779	1.0269
4.995	16.0742	16.5777	1.0313
5.000	16.0656	16.5549	1.0305
5.005	16.0228	16.5587	1.0335
5.010	15.9899	16.5305	1.0338
5.015	15.9549	16.5504	1.0373
5.020	15.9249	16.4984	1.0360
5.025	15.8513	16.4735	1.0393
5.030	15.8005	16.4337	1.0401
5.035	15.8220	16.4377	1.0389
5.040	15.7831	16.3914	1.0385
5.045	15.7965	16.3460	1.0348
5.050	15.7689	16.2749	1.0321
5.055	15.8150	16.2892	1.0300
5.060	15.7842	16.2650	1.0305
5.065	15.8042	16.2234	1.0265
5.070	15.7918	16.2674	1.0301
5.075	15.7889	16.2466	1.0290
5.080	15.7666	16.2220	1.0289
5.085	15.7599	16.2059	1.0283
5.090	15.7722	16.2372	1.0295
5.095	15.7885	16.2622	1.0300
5.100	15.7743	16.2566	1.0306
5.105	15.7483	16.3253	1.0366
5.110	15.7253	16.2370	1.0325
5.115	15.6988	16.2309	1.0339
5.120	15.6865	16.2319	1.0348
5.125	15.6753	16.2412	1.0361
5.130	15.6625	16.2115	1.0350
5.135	15.6140	16.2007	1.0376
5.140	15.6090	16.2228	1.0393
5.145	15.5821	16.1577	1.0369

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	15.5721	16.2042	1.0406
5.155	15.5553	16.1668	1.0393
5.160	15.5422	16.1533	1.0393
5.165	15.5150	16.1480	1.0408
5.170	15.5065	16.1202	1.0396
5.175	15.4855	16.1014	1.0398
5.180	15.4850	16.0986	1.0396
5.185	15.4547	16.0920	1.0412
5.190	15.4653	16.0795	1.0397
5.195	15.4601	16.0698	1.0394
5.200	15.4569	16.0720	1.0398
5.205	15.4550	16.0759	1.0402
5.210	15.4463	16.0614	1.0398
5.215	15.4420	16.0635	1.0402
5.220	15.4186	16.0448	1.0406
5.225	15.3912	16.0414	1.0422
5.230	15.4070	16.0395	1.0411
5.235	15.3695	16.0518	1.0444
5.240	15.3546	16.0282	1.0439
5.245	15.3590	16.0425	1.0445
5.250	15.3293	16.0501	1.0470
5.255	15.2896	16.0475	1.0496
5.260	15.3467	16.0219	1.0440
5.265	15.3393	16.0643	1.0473
5.270	15.2940	16.0267	1.0479
5.275	15.2731	15.9918	1.0471
5.280	15.2774	16.0245	1.0489
5.285	15.2859	15.9971	1.0465
5.290	15.2493	15.9721	1.0474
5.295	15.2940	15.9322	1.0417
5.300	15.2344	15.9634	1.0478
5.305	15.2599	15.9686	1.0464
5.310	15.1624	15.9450	1.0516
5.315	15.2009	15.9549	1.0496
5.320	15.1598	15.9012	1.0489
5.325	15.1788	15.9305	1.0495
5.330	15.1591	15.9253	1.0505
5.335	15.0926	15.9200	1.0548
5.340	15.0834	15.9507	1.0575
5.345	15.0581	15.9716	1.0607
5.350	15.1252	15.9017	1.0513
5.355	15.0473	15.9150	1.0577
5.360	15.0467	15.8358	1.0524
5.365	15.1188	15.9033	1.0519
5.370	15.0160	15.8378	1.0547
5.375	15.0357	15.8902	1.0568

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	15.0411	15.8926	1.0566
5.385	14.9909	15.8513	1.0574
5.390	14.9817	15.9061	1.0617
5.395	15.0252	15.8978	1.0581
5.400	14.9725	15.8758	1.0603
5.405	14.9753	15.8630	1.0593
5.410	14.9522	15.8727	1.0616
5.415	14.9577	15.8268	1.0581
5.420	14.9145	15.8681	1.0639
5.425	14.9033	15.8735	1.0651
5.430	14.9302	15.8321	1.0604
5.435	14.9138	15.8227	1.0609
5.440	14.8531	15.8506	1.0672
5.445	14.9035	15.8418	1.0630
5.450	14.8424	15.8215	1.0660
5.455	14.8717	15.8338	1.0647
5.460	14.8480	15.8230	1.0657
5.465	14.8588	15.8627	1.0676
5.470	14.7843	15.8247	1.0704
5.475	14.8148	15.8100	1.0672
5.480	14.7866	15.8340	1.0708
5.485	14.7996	15.7989	1.0675
5.490	14.7399	15.8392	1.0746
5.495	14.7909	15.8000	1.0682
5.500	14.7651	15.8255	1.0718
5.505	14.7629	15.8280	1.0722
5.510	14.7630	15.8133	1.0711
5.515	14.7652	15.8000	1.0701
5.520	14.7313	15.7887	1.0718
5.525	14.7493	15.7874	1.0704
5.530	14.7338	15.7865	1.0714
5.535	14.7440	15.7688	1.0695
5.540	14.7148	15.7815	1.0725
5.545	14.6915	15.7537	1.0723
5.550	14.7020	15.7549	1.0716
5.555	14.7021	15.7411	1.0707
5.560	14.6890	15.7413	1.0716
5.565	14.6989	15.7399	1.0708
5.570	14.6904	15.7401	1.0715
5.575	14.6951	15.7342	1.0707
5.580	14.6825	15.7557	1.0731
5.585	14.6779	15.7379	1.0722
5.590	14.6876	15.7198	1.0703
5.595	14.6709	15.7120	1.0710
5.600	14.6775	15.7138	1.0706
5.605	14.6944	15.6848	1.0674

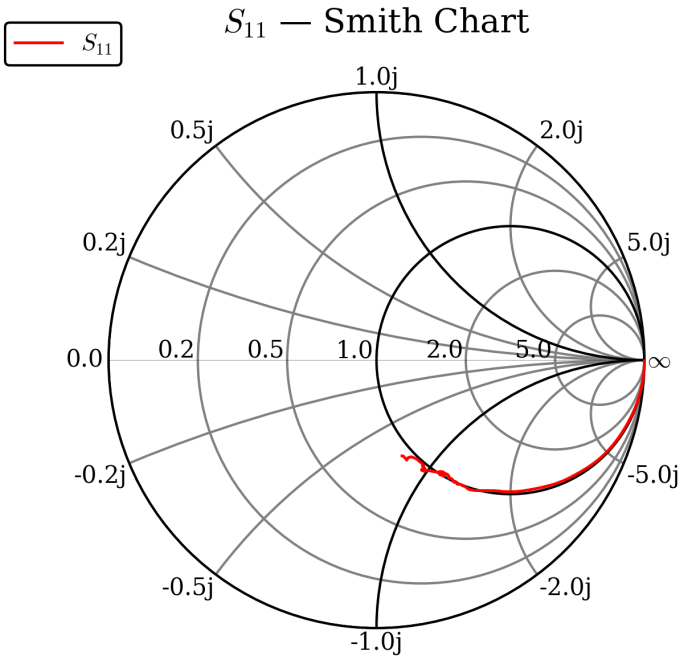
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	14.6631	15.6611	1.0681
5.615	14.6538	15.6290	1.0666
5.620	14.6425	15.6334	1.0677
5.625	14.6376	15.6401	1.0685
5.630	14.6133	15.6010	1.0676
5.635	14.6161	15.5859	1.0664
5.640	14.6043	15.5491	1.0647
5.645	14.5683	15.5355	1.0664
5.650	14.5732	15.5748	1.0687
5.655	14.5415	15.5474	1.0692
5.660	14.5216	15.5496	1.0708
5.665	14.5841	15.5407	1.0656
5.670	14.4539	15.5455	1.0755
5.675	14.4961	15.4904	1.0686
5.680	14.4673	15.4969	1.0712
5.685	14.5146	15.4725	1.0660
5.690	14.4769	15.4666	1.0684
5.695	14.4555	15.5050	1.0726
5.700	14.4320	15.4255	1.0688
5.705	14.3717	15.4278	1.0735
5.710	14.4344	15.4138	1.0678
5.715	14.3859	15.3724	1.0686
5.720	14.3777	15.3940	1.0707
5.725	14.3351	15.3948	1.0739
5.730	14.3385	15.3714	1.0720
5.735	14.3247	15.3680	1.0728
5.740	14.3083	15.3772	1.0747
5.745	14.3022	15.3944	1.0764
5.750	14.2641	15.3512	1.0762
5.755	14.2761	15.3525	1.0754
5.760	14.2259	15.3162	1.0766
5.765	14.2714	15.3117	1.0729
5.770	14.2443	15.2852	1.0731
5.775	14.2384	15.3097	1.0752
5.780	14.1833	15.3079	1.0793
5.785	14.1705	15.2845	1.0786
5.790	14.1410	15.2995	1.0819
5.795	14.1496	15.3147	1.0823
5.800	14.1223	15.2914	1.0828
5.805	14.1158	15.2653	1.0814
5.810	14.1056	15.2908	1.0840
5.815	14.1031	15.2901	1.0842
5.820	14.0912	15.2647	1.0833
5.825	14.0748	15.2633	1.0844
5.830	14.0479	15.2538	1.0858
5.835	14.0481	15.2519	1.0857

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	14.0319	15.2459	1.0865
5.845	14.0163	15.2570	1.0885
5.850	14.0096	15.2253	1.0868
5.855	13.9975	15.2220	1.0875
5.860	13.9702	15.2185	1.0894
5.865	13.9758	15.2306	1.0898
5.870	13.9604	15.1958	1.0885
5.875	13.9514	15.1961	1.0892
5.880	13.9410	15.1881	1.0895
5.885	13.9561	15.1579	1.0861
5.890	13.8897	15.1932	1.0938
5.895	13.8929	15.1647	1.0915
5.900	13.8980	15.1246	1.0883
5.905	13.8843	15.1766	1.0931
5.910	13.8589	15.1034	1.0898
5.915	13.8501	15.1310	1.0925
5.920	13.8434	15.1148	1.0918
5.925	13.8320	15.1003	1.0917
5.930	13.7725	15.1188	1.0978
5.935	13.7651	15.0990	1.0969
5.940	13.7615	15.0993	1.0972
5.945	13.8128	15.0806	1.0918
5.950	13.7551	15.0977	1.0976
5.955	13.7555	15.1012	1.0978
5.960	13.7362	15.0828	1.0980
5.965	13.7250	15.0809	1.0988
5.970	13.7113	15.0997	1.1013
5.975	13.7481	15.0052	1.0914
5.980	13.8095	15.0689	1.0912
5.985	13.6686	15.0736	1.1028
5.990	13.6679	15.0242	1.0992
5.995	13.6754	15.0102	1.0976
6.000	13.7124	15.0056	1.0943

1.2 S_{11} — Smith Chart



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0000	-0.0028	-0.00
0.010	0.9998	-0.0056	-0.00
0.015	0.9999	-0.0085	-0.00
0.020	0.9999	-0.0114	-0.00
0.025	0.9997	-0.0142	-0.00
0.030	0.9997	-0.0171	-0.00
0.035	0.9996	-0.0199	-0.00
0.040	0.9993	-0.0230	-0.00
0.045	0.9992	-0.0258	-0.00
0.050	0.9991	-0.0285	-0.00
0.055	0.9988	-0.0313	-0.01
0.060	0.9987	-0.0343	-0.01
0.065	0.9985	-0.0372	-0.01
0.070	0.9982	-0.0401	-0.01
0.075	0.9980	-0.0428	-0.01
0.080	0.9977	-0.0457	-0.01
0.085	0.9974	-0.0485	-0.01
0.090	0.9971	-0.0515	-0.01
0.095	0.9968	-0.0542	-0.02
0.100	0.9965	-0.0571	-0.02
0.105	0.9961	-0.0599	-0.02
0.110	0.9957	-0.0627	-0.02
0.115	0.9954	-0.0656	-0.02

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9949	-0.0683	-0.02
0.125	0.9947	-0.0712	-0.02
0.130	0.9941	-0.0740	-0.03
0.135	0.9937	-0.0768	-0.03
0.140	0.9932	-0.0796	-0.03
0.145	0.9926	-0.0825	-0.03
0.150	0.9922	-0.0852	-0.04
0.155	0.9916	-0.0880	-0.04
0.160	0.9911	-0.0907	-0.04
0.165	0.9906	-0.0935	-0.04
0.170	0.9901	-0.0963	-0.05
0.175	0.9894	-0.0990	-0.05
0.180	0.9889	-0.1018	-0.05
0.185	0.9882	-0.1045	-0.05
0.190	0.9877	-0.1074	-0.06
0.195	0.9870	-0.1102	-0.06
0.200	0.9864	-0.1128	-0.06
0.205	0.9857	-0.1156	-0.07
0.210	0.9850	-0.1183	-0.07
0.215	0.9842	-0.1210	-0.07
0.220	0.9835	-0.1239	-0.08
0.225	0.9826	-0.1267	-0.08
0.230	0.9820	-0.1293	-0.08
0.235	0.9811	-0.1320	-0.09
0.240	0.9804	-0.1348	-0.09
0.245	0.9795	-0.1375	-0.09
0.250	0.9787	-0.1401	-0.10
0.255	0.9778	-0.1429	-0.10
0.260	0.9770	-0.1456	-0.11
0.265	0.9761	-0.1483	-0.11
0.270	0.9752	-0.1509	-0.12
0.275	0.9743	-0.1535	-0.12
0.280	0.9734	-0.1563	-0.12
0.285	0.9725	-0.1589	-0.13
0.290	0.9716	-0.1614	-0.13
0.295	0.9706	-0.1641	-0.14
0.300	0.9696	-0.1667	-0.14
0.305	0.9686	-0.1693	-0.15
0.310	0.9675	-0.1719	-0.15
0.315	0.9665	-0.1745	-0.16
0.320	0.9655	-0.1770	-0.16
0.325	0.9644	-0.1796	-0.17
0.330	0.9633	-0.1822	-0.17
0.335	0.9622	-0.1846	-0.18
0.340	0.9611	-0.1872	-0.18
0.345	0.9601	-0.1898	-0.19

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9589	-0.1923	-0.19
0.355	0.9579	-0.1948	-0.20
0.360	0.9569	-0.1973	-0.20
0.365	0.9556	-0.1998	-0.21
0.370	0.9544	-0.2022	-0.21
0.375	0.9532	-0.2048	-0.22
0.380	0.9520	-0.2072	-0.23
0.385	0.9507	-0.2096	-0.23
0.390	0.9496	-0.2121	-0.24
0.395	0.9483	-0.2144	-0.24
0.400	0.9472	-0.2168	-0.25
0.405	0.9460	-0.2193	-0.26
0.410	0.9447	-0.2216	-0.26
0.415	0.9437	-0.2239	-0.27
0.420	0.9423	-0.2262	-0.27
0.425	0.9412	-0.2286	-0.28
0.430	0.9399	-0.2309	-0.28
0.435	0.9387	-0.2331	-0.29
0.440	0.9374	-0.2355	-0.30
0.445	0.9360	-0.2377	-0.30
0.450	0.9349	-0.2399	-0.31
0.455	0.9336	-0.2421	-0.31
0.460	0.9323	-0.2444	-0.32
0.465	0.9309	-0.2466	-0.33
0.470	0.9296	-0.2489	-0.33
0.475	0.9281	-0.2510	-0.34
0.480	0.9270	-0.2532	-0.35
0.485	0.9256	-0.2553	-0.35
0.490	0.9243	-0.2575	-0.36
0.495	0.9231	-0.2596	-0.36
0.500	0.9217	-0.2618	-0.37
0.505	0.9203	-0.2638	-0.38
0.510	0.9190	-0.2660	-0.38
0.515	0.9176	-0.2681	-0.39
0.520	0.9164	-0.2701	-0.40
0.525	0.9150	-0.2722	-0.40
0.530	0.9138	-0.2741	-0.41
0.535	0.9125	-0.2761	-0.42
0.540	0.9111	-0.2783	-0.42
0.545	0.9097	-0.2803	-0.43
0.550	0.9083	-0.2823	-0.43
0.555	0.9069	-0.2843	-0.44
0.560	0.9055	-0.2863	-0.45
0.565	0.9041	-0.2882	-0.45
0.570	0.9028	-0.2901	-0.46
0.575	0.9014	-0.2922	-0.47

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8999	-0.2941	-0.48
0.585	0.8986	-0.2961	-0.48
0.590	0.8970	-0.2980	-0.49
0.595	0.8957	-0.2999	-0.50
0.600	0.8943	-0.3017	-0.50
0.605	0.8928	-0.3036	-0.51
0.610	0.8914	-0.3053	-0.52
0.615	0.8899	-0.3072	-0.52
0.620	0.8884	-0.3090	-0.53
0.625	0.8871	-0.3108	-0.54
0.630	0.8856	-0.3126	-0.55
0.635	0.8841	-0.3145	-0.55
0.640	0.8828	-0.3162	-0.56
0.645	0.8814	-0.3180	-0.57
0.650	0.8798	-0.3198	-0.57
0.655	0.8784	-0.3216	-0.58
0.660	0.8770	-0.3233	-0.59
0.665	0.8755	-0.3250	-0.59
0.670	0.8740	-0.3266	-0.60
0.675	0.8725	-0.3283	-0.61
0.680	0.8710	-0.3300	-0.62
0.685	0.8696	-0.3317	-0.62
0.690	0.8681	-0.3332	-0.63
0.695	0.8665	-0.3349	-0.64
0.700	0.8649	-0.3366	-0.65
0.705	0.8635	-0.3382	-0.65
0.710	0.8619	-0.3398	-0.66
0.715	0.8604	-0.3415	-0.67
0.720	0.8588	-0.3430	-0.68
0.725	0.8572	-0.3446	-0.69
0.730	0.8557	-0.3463	-0.70
0.735	0.8540	-0.3479	-0.70
0.740	0.8524	-0.3495	-0.71
0.745	0.8507	-0.3510	-0.72
0.750	0.8493	-0.3524	-0.73
0.755	0.8477	-0.3539	-0.74
0.760	0.8460	-0.3555	-0.75
0.765	0.8444	-0.3570	-0.75
0.770	0.8427	-0.3585	-0.76
0.775	0.8409	-0.3601	-0.77
0.780	0.8394	-0.3616	-0.78
0.785	0.8378	-0.3632	-0.79
0.790	0.8360	-0.3648	-0.80
0.795	0.8344	-0.3662	-0.81
0.800	0.8327	-0.3675	-0.82
0.805	0.8309	-0.3690	-0.83

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.810	0.8292	-0.3705	-0.84
0.815	0.8275	-0.3719	-0.85
0.820	0.8257	-0.3733	-0.86
0.825	0.8240	-0.3748	-0.86
0.830	0.8223	-0.3762	-0.87
0.835	0.8203	-0.3776	-0.89
0.840	0.8188	-0.3789	-0.89
0.845	0.8170	-0.3803	-0.90
0.850	0.8152	-0.3816	-0.91
0.855	0.8133	-0.3829	-0.93
0.860	0.8117	-0.3844	-0.93
0.865	0.8098	-0.3856	-0.95
0.870	0.8080	-0.3870	-0.96
0.875	0.8061	-0.3882	-0.97
0.880	0.8044	-0.3896	-0.98
0.885	0.8025	-0.3908	-0.99
0.890	0.8006	-0.3920	-1.00
0.895	0.7989	-0.3933	-1.01
0.900	0.7969	-0.3946	-1.02
0.905	0.7952	-0.3959	-1.03
0.910	0.7932	-0.3970	-1.04
0.915	0.7915	-0.3983	-1.05
0.920	0.7895	-0.3995	-1.06
0.925	0.7877	-0.4007	-1.07
0.930	0.7858	-0.4019	-1.08
0.935	0.7841	-0.4032	-1.09
0.940	0.7821	-0.4043	-1.11
0.945	0.7804	-0.4054	-1.12
0.950	0.7785	-0.4067	-1.13
0.955	0.7766	-0.4078	-1.14
0.960	0.7748	-0.4089	-1.15
0.965	0.7729	-0.4101	-1.16
0.970	0.7710	-0.4113	-1.17
0.975	0.7692	-0.4123	-1.18
0.980	0.7675	-0.4134	-1.19
0.985	0.7656	-0.4144	-1.20
0.990	0.7637	-0.4155	-1.22
0.995	0.7619	-0.4166	-1.23
1.000	0.7600	-0.4176	-1.24
1.005	0.7583	-0.4186	-1.25
1.010	0.7565	-0.4196	-1.26
1.015	0.7547	-0.4207	-1.27
1.020	0.7527	-0.4216	-1.28
1.025	0.7510	-0.4226	-1.29
1.030	0.7493	-0.4237	-1.30
1.035	0.7474	-0.4248	-1.31

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.040	0.7458	-0.4257	-1.32
1.045	0.7440	-0.4267	-1.33
1.050	0.7422	-0.4276	-1.35
1.055	0.7405	-0.4286	-1.36
1.060	0.7388	-0.4295	-1.37
1.065	0.7371	-0.4304	-1.38
1.070	0.7352	-0.4314	-1.39
1.075	0.7336	-0.4323	-1.40
1.080	0.7319	-0.4333	-1.41
1.085	0.7302	-0.4342	-1.42
1.090	0.7284	-0.4350	-1.43
1.095	0.7267	-0.4358	-1.44
1.100	0.7251	-0.4367	-1.45
1.105	0.7234	-0.4376	-1.46
1.110	0.7216	-0.4385	-1.47
1.115	0.7200	-0.4393	-1.48
1.120	0.7183	-0.4401	-1.49
1.125	0.7167	-0.4409	-1.50
1.130	0.7151	-0.4418	-1.51
1.135	0.7133	-0.4425	-1.52
1.140	0.7118	-0.4433	-1.53
1.145	0.7102	-0.4441	-1.54
1.150	0.7086	-0.4449	-1.55
1.155	0.7069	-0.4457	-1.56
1.160	0.7053	-0.4464	-1.57
1.165	0.7038	-0.4472	-1.58
1.170	0.7021	-0.4478	-1.59
1.175	0.7007	-0.4487	-1.60
1.180	0.6991	-0.4492	-1.61
1.185	0.6975	-0.4499	-1.62
1.190	0.6961	-0.4506	-1.63
1.195	0.6947	-0.4512	-1.64
1.200	0.6932	-0.4518	-1.65
1.205	0.6916	-0.4523	-1.66
1.210	0.6902	-0.4530	-1.66
1.215	0.6887	-0.4535	-1.68
1.220	0.6873	-0.4540	-1.68
1.225	0.6857	-0.4547	-1.69
1.230	0.6843	-0.4552	-1.70
1.235	0.6829	-0.4558	-1.71
1.240	0.6813	-0.4562	-1.72
1.245	0.6798	-0.4570	-1.73
1.250	0.6784	-0.4574	-1.74
1.255	0.6771	-0.4578	-1.75
1.260	0.6757	-0.4584	-1.76
1.265	0.6743	-0.4589	-1.77

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.6730	-0.4594	-1.78
1.275	0.6715	-0.4598	-1.79
1.280	0.6700	-0.4604	-1.80
1.285	0.6687	-0.4609	-1.81
1.290	0.6672	-0.4613	-1.82
1.295	0.6657	-0.4619	-1.83
1.300	0.6644	-0.4624	-1.84
1.305	0.6632	-0.4628	-1.84
1.310	0.6617	-0.4632	-1.86
1.315	0.6603	-0.4637	-1.86
1.320	0.6590	-0.4642	-1.87
1.325	0.6576	-0.4643	-1.88
1.330	0.6563	-0.4648	-1.89
1.335	0.6551	-0.4652	-1.90
1.340	0.6537	-0.4656	-1.91
1.345	0.6523	-0.4659	-1.92
1.350	0.6509	-0.4663	-1.93
1.355	0.6496	-0.4667	-1.94
1.360	0.6484	-0.4670	-1.95
1.365	0.6469	-0.4673	-1.96
1.370	0.6456	-0.4678	-1.97
1.375	0.6442	-0.4681	-1.98
1.380	0.6427	-0.4684	-1.99
1.385	0.6412	-0.4687	-2.00
1.390	0.6400	-0.4693	-2.01
1.395	0.6384	-0.4694	-2.02
1.400	0.6371	-0.4698	-2.03
1.405	0.6356	-0.4700	-2.04
1.410	0.6340	-0.4705	-2.05
1.415	0.6326	-0.4710	-2.06
1.420	0.6311	-0.4712	-2.07
1.425	0.6297	-0.4717	-2.08
1.430	0.6282	-0.4721	-2.09
1.435	0.6266	-0.4724	-2.11
1.440	0.6251	-0.4728	-2.12
1.445	0.6235	-0.4733	-2.13
1.450	0.6221	-0.4736	-2.14
1.455	0.6205	-0.4740	-2.15
1.460	0.6190	-0.4745	-2.16
1.465	0.6175	-0.4747	-2.17
1.470	0.6159	-0.4750	-2.18
1.475	0.6143	-0.4755	-2.19
1.480	0.6127	-0.4758	-2.21
1.485	0.6111	-0.4761	-2.22
1.490	0.6094	-0.4766	-2.23
1.495	0.6078	-0.4769	-2.24

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.500	0.6062	-0.4772	-2.25
1.505	0.6045	-0.4775	-2.27
1.510	0.6028	-0.4778	-2.28
1.515	0.6011	-0.4782	-2.29
1.520	0.5994	-0.4786	-2.30
1.525	0.5977	-0.4790	-2.32
1.530	0.5961	-0.4793	-2.33
1.535	0.5942	-0.4797	-2.34
1.540	0.5926	-0.4800	-2.35
1.545	0.5908	-0.4803	-2.37
1.550	0.5891	-0.4806	-2.38
1.555	0.5872	-0.4809	-2.39
1.560	0.5855	-0.4812	-2.41
1.565	0.5837	-0.4816	-2.42
1.570	0.5818	-0.4819	-2.43
1.575	0.5800	-0.4824	-2.45
1.580	0.5784	-0.4825	-2.46
1.585	0.5765	-0.4829	-2.48
1.590	0.5747	-0.4832	-2.49
1.595	0.5730	-0.4837	-2.50
1.600	0.5711	-0.4840	-2.51
1.605	0.5695	-0.4844	-2.53
1.610	0.5677	-0.4846	-2.54
1.615	0.5658	-0.4848	-2.56
1.620	0.5642	-0.4851	-2.57
1.625	0.5622	-0.4854	-2.58
1.630	0.5605	-0.4858	-2.60
1.635	0.5588	-0.4860	-2.61
1.640	0.5569	-0.4863	-2.62
1.645	0.5549	-0.4866	-2.64
1.650	0.5530	-0.4868	-2.65
1.655	0.5512	-0.4871	-2.67
1.660	0.5496	-0.4874	-2.68
1.665	0.5478	-0.4876	-2.69
1.670	0.5460	-0.4877	-2.71
1.675	0.5443	-0.4881	-2.72
1.680	0.5426	-0.4883	-2.73
1.685	0.5407	-0.4885	-2.75
1.690	0.5391	-0.4887	-2.76
1.695	0.5375	-0.4889	-2.77
1.700	0.5356	-0.4889	-2.79
1.705	0.5340	-0.4891	-2.80
1.710	0.5323	-0.4894	-2.82
1.715	0.5306	-0.4894	-2.83
1.720	0.5288	-0.4897	-2.84
1.725	0.5272	-0.4898	-2.86

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.730	0.5254	-0.4900	-2.87
1.735	0.5237	-0.4902	-2.89
1.740	0.5220	-0.4900	-2.90
1.745	0.5206	-0.4904	-2.91
1.750	0.5189	-0.4906	-2.93
1.755	0.5174	-0.4905	-2.94
1.760	0.5159	-0.4908	-2.95
1.765	0.5146	-0.4909	-2.96
1.770	0.5131	-0.4910	-2.97
1.775	0.5118	-0.4911	-2.98
1.780	0.5104	-0.4911	-3.00
1.785	0.5090	-0.4909	-3.01
1.790	0.5076	-0.4910	-3.02
1.795	0.5062	-0.4911	-3.03
1.800	0.5046	-0.4910	-3.05
1.805	0.5034	-0.4912	-3.06
1.810	0.5020	-0.4913	-3.07
1.815	0.5008	-0.4914	-3.08
1.820	0.4995	-0.4913	-3.09
1.825	0.4983	-0.4915	-3.10
1.830	0.4971	-0.4914	-3.11
1.835	0.4959	-0.4914	-3.12
1.840	0.4947	-0.4914	-3.13
1.845	0.4935	-0.4912	-3.14
1.850	0.4923	-0.4911	-3.16
1.855	0.4911	-0.4910	-3.17
1.860	0.4900	-0.4909	-3.18
1.865	0.4888	-0.4908	-3.19
1.870	0.4877	-0.4909	-3.20
1.875	0.4863	-0.4909	-3.21
1.880	0.4853	-0.4910	-3.22
1.885	0.4844	-0.4911	-3.23
1.890	0.4837	-0.4910	-3.23
1.895	0.4827	-0.4910	-3.24
1.900	0.4817	-0.4908	-3.25
1.905	0.4807	-0.4907	-3.26
1.910	0.4797	-0.4907	-3.27
1.915	0.4787	-0.4908	-3.28
1.920	0.4779	-0.4908	-3.29
1.925	0.4769	-0.4907	-3.30
1.930	0.4761	-0.4907	-3.30
1.935	0.4753	-0.4906	-3.31
1.940	0.4743	-0.4906	-3.32
1.945	0.4736	-0.4906	-3.33
1.950	0.4727	-0.4905	-3.33
1.955	0.4720	-0.4904	-3.34

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.960	0.4712	-0.4905	-3.35
1.965	0.4704	-0.4903	-3.36
1.970	0.4697	-0.4901	-3.36
1.975	0.4689	-0.4900	-3.37
1.980	0.4681	-0.4899	-3.38
1.985	0.4674	-0.4897	-3.39
1.990	0.4665	-0.4897	-3.40
1.995	0.4657	-0.4896	-3.40
2.000	0.4650	-0.4895	-3.41
2.005	0.4642	-0.4894	-3.42
2.010	0.4634	-0.4894	-3.43
2.015	0.4627	-0.4892	-3.44
2.020	0.4618	-0.4891	-3.44
2.025	0.4611	-0.4891	-3.45
2.030	0.4603	-0.4891	-3.46
2.035	0.4595	-0.4888	-3.47
2.040	0.4587	-0.4887	-3.48
2.045	0.4579	-0.4887	-3.48
2.050	0.4571	-0.4886	-3.49
2.055	0.4562	-0.4884	-3.50
2.060	0.4553	-0.4885	-3.51
2.065	0.4545	-0.4882	-3.52
2.070	0.4537	-0.4883	-3.52
2.075	0.4527	-0.4881	-3.53
2.080	0.4518	-0.4881	-3.54
2.085	0.4508	-0.4880	-3.55
2.090	0.4499	-0.4878	-3.56
2.095	0.4488	-0.4875	-3.57
2.100	0.4478	-0.4876	-3.58
2.105	0.4469	-0.4875	-3.59
2.110	0.4457	-0.4874	-3.60
2.115	0.4447	-0.4875	-3.61
2.120	0.4437	-0.4875	-3.62
2.125	0.4426	-0.4874	-3.63
2.130	0.4414	-0.4874	-3.64
2.135	0.4403	-0.4873	-3.65
2.140	0.4392	-0.4875	-3.66
2.145	0.4382	-0.4874	-3.67
2.150	0.4369	-0.4874	-3.68
2.155	0.4357	-0.4873	-3.69
2.160	0.4346	-0.4872	-3.70
2.165	0.4333	-0.4872	-3.71
2.170	0.4321	-0.4872	-3.72
2.175	0.4309	-0.4872	-3.74
2.180	0.4297	-0.4872	-3.75
2.185	0.4284	-0.4871	-3.76

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.190	0.4271	-0.4870	-3.77
2.195	0.4258	-0.4871	-3.78
2.200	0.4245	-0.4871	-3.79
2.205	0.4231	-0.4870	-3.81
2.210	0.4218	-0.4870	-3.82
2.215	0.4204	-0.4870	-3.83
2.220	0.4192	-0.4869	-3.84
2.225	0.4178	-0.4870	-3.85
2.230	0.4165	-0.4868	-3.87
2.235	0.4151	-0.4869	-3.88
2.240	0.4137	-0.4869	-3.89
2.245	0.4124	-0.4869	-3.90
2.250	0.4109	-0.4869	-3.92
2.255	0.4095	-0.4869	-3.93
2.260	0.4083	-0.4869	-3.94
2.265	0.4069	-0.4870	-3.95
2.270	0.4055	-0.4869	-3.96
2.275	0.4042	-0.4870	-3.97
2.280	0.4027	-0.4870	-3.99
2.285	0.4014	-0.4869	-4.00
2.290	0.4000	-0.4870	-4.01
2.295	0.3987	-0.4870	-4.02
2.300	0.3973	-0.4869	-4.03
2.305	0.3959	-0.4870	-4.05
2.310	0.3946	-0.4870	-4.06
2.315	0.3931	-0.4869	-4.07
2.320	0.3917	-0.4870	-4.08
2.325	0.3905	-0.4869	-4.09
2.330	0.3891	-0.4872	-4.10
2.335	0.3878	-0.4870	-4.12
2.340	0.3864	-0.4871	-4.13
2.345	0.3850	-0.4870	-4.14
2.350	0.3837	-0.4870	-4.15
2.355	0.3825	-0.4870	-4.16
2.360	0.3813	-0.4869	-4.17
2.365	0.3800	-0.4869	-4.19
2.370	0.3787	-0.4868	-4.20
2.375	0.3776	-0.4866	-4.21
2.380	0.3764	-0.4867	-4.22
2.385	0.3752	-0.4867	-4.23
2.390	0.3741	-0.4866	-4.24
2.395	0.3729	-0.4866	-4.25
2.400	0.3718	-0.4865	-4.26
2.405	0.3707	-0.4865	-4.27
2.410	0.3696	-0.4865	-4.28
2.415	0.3686	-0.4865	-4.29

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.420	0.3675	-0.4864	-4.30
2.425	0.3665	-0.4863	-4.31
2.430	0.3655	-0.4862	-4.32
2.435	0.3644	-0.4860	-4.33
2.440	0.3635	-0.4860	-4.34
2.445	0.3625	-0.4859	-4.35
2.450	0.3618	-0.4857	-4.36
2.455	0.3608	-0.4855	-4.37
2.460	0.3600	-0.4853	-4.38
2.465	0.3590	-0.4852	-4.39
2.470	0.3582	-0.4850	-4.39
2.475	0.3574	-0.4849	-4.40
2.480	0.3566	-0.4847	-4.41
2.485	0.3557	-0.4847	-4.42
2.490	0.3549	-0.4845	-4.43
2.495	0.3541	-0.4843	-4.44
2.500	0.3533	-0.4842	-4.45
2.505	0.3526	-0.4842	-4.45
2.510	0.3518	-0.4840	-4.46
2.515	0.3511	-0.4838	-4.47
2.520	0.3503	-0.4838	-4.48
2.525	0.3494	-0.4836	-4.49
2.530	0.3487	-0.4835	-4.49
2.535	0.3480	-0.4835	-4.50
2.540	0.3473	-0.4834	-4.51
2.545	0.3466	-0.4833	-4.51
2.550	0.3460	-0.4832	-4.52
2.555	0.3454	-0.4828	-4.53
2.560	0.3448	-0.4825	-4.54
2.565	0.3444	-0.4821	-4.55
2.570	0.3440	-0.4816	-4.56
2.575	0.3439	-0.4812	-4.56
2.580	0.3437	-0.4807	-4.57
2.585	0.3437	-0.4805	-4.57
2.590	0.3435	-0.4804	-4.57
2.595	0.3435	-0.4804	-4.57
2.600	0.3433	-0.4803	-4.58
2.605	0.3432	-0.4803	-4.58
2.610	0.3432	-0.4803	-4.58
2.615	0.3428	-0.4801	-4.58
2.620	0.3426	-0.4801	-4.59
2.625	0.3423	-0.4801	-4.59
2.630	0.3421	-0.4799	-4.59
2.635	0.3418	-0.4798	-4.60
2.640	0.3416	-0.4795	-4.60
2.645	0.3414	-0.4794	-4.60

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	0.3411	-0.4793	-4.61
2.655	0.3408	-0.4791	-4.61
2.660	0.3405	-0.4789	-4.62
2.665	0.3401	-0.4787	-4.62
2.670	0.3397	-0.4787	-4.63
2.675	0.3393	-0.4786	-4.63
2.680	0.3390	-0.4783	-4.64
2.685	0.3387	-0.4783	-4.64
2.690	0.3383	-0.4784	-4.64
2.695	0.3380	-0.4781	-4.65
2.700	0.3377	-0.4780	-4.65
2.705	0.3375	-0.4779	-4.66
2.710	0.3373	-0.4778	-4.66
2.715	0.3370	-0.4775	-4.66
2.720	0.3368	-0.4771	-4.67
2.725	0.3365	-0.4768	-4.68
2.730	0.3363	-0.4766	-4.68
2.735	0.3360	-0.4762	-4.69
2.740	0.3356	-0.4756	-4.70
2.745	0.3352	-0.4753	-4.71
2.750	0.3348	-0.4748	-4.72
2.755	0.3343	-0.4745	-4.73
2.760	0.3338	-0.4742	-4.73
2.765	0.3332	-0.4738	-4.74
2.770	0.3328	-0.4733	-4.75
2.775	0.3322	-0.4729	-4.76
2.780	0.3316	-0.4725	-4.77
2.785	0.3309	-0.4722	-4.78
2.790	0.3302	-0.4719	-4.79
2.795	0.3296	-0.4715	-4.80
2.800	0.3289	-0.4712	-4.81
2.805	0.3283	-0.4709	-4.82
2.810	0.3277	-0.4706	-4.83
2.815	0.3270	-0.4704	-4.84
2.820	0.3263	-0.4702	-4.85
2.825	0.3256	-0.4699	-4.86
2.830	0.3251	-0.4697	-4.86
2.835	0.3244	-0.4695	-4.87
2.840	0.3237	-0.4694	-4.88
2.845	0.3230	-0.4691	-4.89
2.850	0.3224	-0.4689	-4.90
2.855	0.3219	-0.4686	-4.90
2.860	0.3212	-0.4684	-4.91
2.865	0.3207	-0.4683	-4.92
2.870	0.3202	-0.4680	-4.93
2.875	0.3197	-0.4676	-4.94

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.880	0.3192	-0.4672	-4.95
2.885	0.3187	-0.4666	-4.96
2.890	0.3180	-0.4663	-4.97
2.895	0.3173	-0.4656	-4.98
2.900	0.3165	-0.4651	-5.00
2.905	0.3157	-0.4646	-5.01
2.910	0.3147	-0.4642	-5.02
2.915	0.3138	-0.4639	-5.04
2.920	0.3129	-0.4636	-5.05
2.925	0.3120	-0.4632	-5.06
2.930	0.3109	-0.4631	-5.07
2.935	0.3100	-0.4630	-5.08
2.940	0.3092	-0.4629	-5.09
2.945	0.3084	-0.4630	-5.09
2.950	0.3076	-0.4629	-5.10
2.955	0.3069	-0.4629	-5.11
2.960	0.3063	-0.4629	-5.11
2.965	0.3058	-0.4629	-5.12
2.970	0.3054	-0.4631	-5.12
2.975	0.3048	-0.4631	-5.12
2.980	0.3043	-0.4631	-5.13
2.985	0.3041	-0.4630	-5.13
2.990	0.3039	-0.4631	-5.13
2.995	0.3039	-0.4631	-5.13
3.000	0.3036	-0.4628	-5.14
3.005	0.3015	-0.4623	-5.16
3.010	0.3012	-0.4626	-5.16
3.015	0.3011	-0.4626	-5.16
3.020	0.3010	-0.4628	-5.16
3.025	0.3011	-0.4628	-5.16
3.030	0.3011	-0.4627	-5.16
3.035	0.3012	-0.4622	-5.17
3.040	0.3012	-0.4618	-5.17
3.045	0.3013	-0.4612	-5.18
3.050	0.3014	-0.4608	-5.18
3.055	0.3014	-0.4601	-5.19
3.060	0.3012	-0.4594	-5.20
3.065	0.3010	-0.4587	-5.21
3.070	0.3005	-0.4579	-5.23
3.075	0.3001	-0.4572	-5.24
3.080	0.2998	-0.4563	-5.26
3.085	0.2992	-0.4558	-5.27
3.090	0.2984	-0.4552	-5.28
3.095	0.2976	-0.4548	-5.30
3.100	0.2968	-0.4545	-5.31
3.105	0.2961	-0.4544	-5.31

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	0.2954	-0.4542	-5.32
3.115	0.2946	-0.4539	-5.33
3.120	0.2939	-0.4540	-5.34
3.125	0.2933	-0.4543	-5.34
3.130	0.2927	-0.4544	-5.34
3.135	0.2922	-0.4546	-5.35
3.140	0.2919	-0.4551	-5.34
3.145	0.2916	-0.4555	-5.34
3.150	0.2913	-0.4557	-5.34
3.155	0.2913	-0.4560	-5.33
3.160	0.2913	-0.4562	-5.33
3.165	0.2914	-0.4565	-5.33
3.170	0.2915	-0.4567	-5.32
3.175	0.2916	-0.4571	-5.32
3.180	0.2920	-0.4570	-5.32
3.185	0.2922	-0.4568	-5.32
3.190	0.2927	-0.4565	-5.32
3.195	0.2927	-0.4560	-5.32
3.200	0.2930	-0.4553	-5.33
3.205	0.2930	-0.4547	-5.34
3.210	0.2929	-0.4540	-5.35
3.215	0.2928	-0.4533	-5.36
3.220	0.2923	-0.4523	-5.38
3.225	0.2918	-0.4519	-5.39
3.230	0.2910	-0.4517	-5.40
3.235	0.2904	-0.4516	-5.40
3.240	0.2900	-0.4515	-5.41
3.245	0.2898	-0.4518	-5.40
3.250	0.2893	-0.4518	-5.41
3.255	0.2892	-0.4518	-5.41
3.260	0.2887	-0.4522	-5.41
3.265	0.2884	-0.4521	-5.41
3.270	0.2881	-0.4523	-5.41
3.275	0.2881	-0.4524	-5.41
3.280	0.2880	-0.4524	-5.41
3.285	0.2876	-0.4525	-5.41
3.290	0.2872	-0.4527	-5.42
3.295	0.2872	-0.4528	-5.41
3.300	0.2871	-0.4527	-5.42
3.305	0.2870	-0.4527	-5.42
3.310	0.2868	-0.4530	-5.41
3.315	0.2866	-0.4528	-5.42
3.320	0.2867	-0.4526	-5.42
3.325	0.2866	-0.4528	-5.42
3.330	0.2865	-0.4526	-5.42
3.335	0.2865	-0.4524	-5.43

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	0.2865	-0.4523	-5.43
3.345	0.2864	-0.4520	-5.43
3.350	0.2863	-0.4516	-5.44
3.355	0.2863	-0.4513	-5.44
3.360	0.2861	-0.4509	-5.45
3.365	0.2862	-0.4505	-5.45
3.370	0.2859	-0.4500	-5.46
3.375	0.2856	-0.4494	-5.47
3.380	0.2854	-0.4489	-5.48
3.385	0.2851	-0.4484	-5.49
3.390	0.2846	-0.4477	-5.51
3.395	0.2842	-0.4472	-5.52
3.400	0.2838	-0.4465	-5.53
3.405	0.2835	-0.4457	-5.54
3.410	0.2828	-0.4450	-5.56
3.415	0.2822	-0.4444	-5.57
3.420	0.2816	-0.4440	-5.58
3.425	0.2808	-0.4432	-5.60
3.430	0.2802	-0.4427	-5.61
3.435	0.2793	-0.4422	-5.63
3.440	0.2784	-0.4418	-5.64
3.445	0.2773	-0.4411	-5.66
3.450	0.2765	-0.4408	-5.67
3.455	0.2755	-0.4407	-5.69
3.460	0.2745	-0.4404	-5.70
3.465	0.2739	-0.4403	-5.70
3.470	0.2730	-0.4402	-5.71
3.475	0.2721	-0.4400	-5.72
3.480	0.2713	-0.4400	-5.73
3.485	0.2705	-0.4400	-5.74
3.490	0.2696	-0.4403	-5.74
3.495	0.2693	-0.4404	-5.74
3.500	0.2691	-0.4407	-5.74
3.505	0.2690	-0.4411	-5.74
3.510	0.2688	-0.4410	-5.74
3.515	0.2687	-0.4409	-5.74
3.520	0.2689	-0.4406	-5.74
3.525	0.2692	-0.4405	-5.74
3.530	0.2692	-0.4401	-5.75
3.535	0.2698	-0.4393	-5.75
3.540	0.2704	-0.4381	-5.77
3.545	0.2702	-0.4365	-5.79
3.550	0.2703	-0.4348	-5.82
3.555	0.2697	-0.4331	-5.84
3.560	0.2688	-0.4312	-5.88
3.565	0.2678	-0.4294	-5.92

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.570	0.2659	-0.4277	-5.96
3.575	0.2639	-0.4260	-6.00
3.580	0.2615	-0.4246	-6.04
3.585	0.2587	-0.4237	-6.08
3.590	0.2561	-0.4245	-6.09
3.595	0.2537	-0.4252	-6.10
3.600	0.2521	-0.4267	-6.10
3.605	0.2507	-0.4284	-6.08
3.610	0.2507	-0.4301	-6.06
3.615	0.2508	-0.4312	-6.04
3.620	0.2516	-0.4319	-6.02
3.625	0.2523	-0.4325	-6.01
3.630	0.2531	-0.4327	-6.00
3.635	0.2544	-0.4322	-5.99
3.640	0.2549	-0.4315	-6.00
3.645	0.2552	-0.4306	-6.01
3.650	0.2554	-0.4294	-6.03
3.655	0.2553	-0.4284	-6.04
3.660	0.2552	-0.4274	-6.06
3.665	0.2550	-0.4264	-6.08
3.670	0.2545	-0.4253	-6.10
3.675	0.2543	-0.4245	-6.11
3.680	0.2541	-0.4234	-6.13
3.685	0.2534	-0.4225	-6.15
3.690	0.2528	-0.4211	-6.18
3.695	0.2520	-0.4204	-6.19
3.700	0.2510	-0.4193	-6.22
3.705	0.2497	-0.4182	-6.25
3.710	0.2481	-0.4174	-6.27
3.715	0.2466	-0.4167	-6.30
3.720	0.2445	-0.4164	-6.32
3.725	0.2425	-0.4166	-6.34
3.730	0.2399	-0.4169	-6.36
3.735	0.2381	-0.4179	-6.36
3.740	0.2364	-0.4196	-6.35
3.745	0.2355	-0.4213	-6.33
3.750	0.2350	-0.4237	-6.29
3.755	0.2354	-0.4256	-6.26
3.760	0.2365	-0.4275	-6.22
3.765	0.2385	-0.4287	-6.19
3.770	0.2412	-0.4290	-6.16
3.775	0.2439	-0.4291	-6.13
3.780	0.2460	-0.4263	-6.16
3.785	0.2464	-0.4233	-6.20
3.790	0.2461	-0.4208	-6.24
3.795	0.2449	-0.4190	-6.28

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	0.2435	-0.4175	-6.32
3.805	0.2416	-0.4165	-6.35
3.810	0.2402	-0.4159	-6.37
3.815	0.2385	-0.4157	-6.39
3.820	0.2371	-0.4156	-6.40
3.825	0.2356	-0.4157	-6.42
3.830	0.2344	-0.4162	-6.42
3.835	0.2339	-0.4189	-6.38
3.840	0.2335	-0.4189	-6.38
3.845	0.2332	-0.4190	-6.38
3.850	0.2328	-0.4189	-6.39
3.855	0.2323	-0.4189	-6.39
3.860	0.2320	-0.4189	-6.40
3.865	0.2315	-0.4187	-6.40
3.870	0.2312	-0.4187	-6.41
3.875	0.2308	-0.4187	-6.41
3.880	0.2302	-0.4185	-6.42
3.885	0.2297	-0.4185	-6.42
3.890	0.2290	-0.4185	-6.43
3.895	0.2285	-0.4182	-6.44
3.900	0.2280	-0.4183	-6.44
3.905	0.2272	-0.4182	-6.45
3.910	0.2265	-0.4182	-6.46
3.915	0.2259	-0.4180	-6.46
3.920	0.2251	-0.4181	-6.47
3.925	0.2244	-0.4179	-6.48
3.930	0.2238	-0.4177	-6.49
3.935	0.2230	-0.4176	-6.49
3.940	0.2224	-0.4176	-6.50
3.945	0.2215	-0.4174	-6.51
3.950	0.2208	-0.4175	-6.52
3.955	0.2200	-0.4173	-6.53
3.960	0.2191	-0.4172	-6.54
3.965	0.2184	-0.4168	-6.55
3.970	0.2176	-0.4168	-6.56
3.975	0.2167	-0.4165	-6.57
3.980	0.2158	-0.4165	-6.58
3.985	0.2148	-0.4164	-6.59
3.990	0.2141	-0.4163	-6.59
3.995	0.2130	-0.4161	-6.61
4.000	0.2120	-0.4160	-6.62
4.005	0.2112	-0.4157	-6.63
4.010	0.2102	-0.4155	-6.64
4.015	0.2092	-0.4153	-6.65
4.020	0.2084	-0.4150	-6.66
4.025	0.2075	-0.4150	-6.67

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	0.2066	-0.4146	-6.68
4.035	0.2055	-0.4145	-6.69
4.040	0.2044	-0.4144	-6.71
4.045	0.2036	-0.4141	-6.72
4.050	0.2026	-0.4138	-6.73
4.055	0.2015	-0.4135	-6.75
4.060	0.2006	-0.4134	-6.75
4.065	0.1995	-0.4132	-6.77
4.070	0.1986	-0.4129	-6.78
4.075	0.1976	-0.4127	-6.79
4.080	0.1967	-0.4125	-6.80
4.085	0.1958	-0.4122	-6.81
4.090	0.1947	-0.4121	-6.83
4.095	0.1937	-0.4120	-6.83
4.100	0.1928	-0.4116	-6.85
4.105	0.1917	-0.4116	-6.86
4.110	0.1909	-0.4115	-6.87
4.115	0.1896	-0.4113	-6.88
4.120	0.1885	-0.4113	-6.89
4.125	0.1876	-0.4114	-6.89
4.130	0.1862	-0.4111	-6.91
4.135	0.1850	-0.4108	-6.93
4.140	0.1839	-0.4107	-6.94
4.145	0.1827	-0.4106	-6.95
4.150	0.1814	-0.4101	-6.97
4.155	0.1805	-0.4096	-6.98
4.160	0.1796	-0.4092	-7.00
4.165	0.1788	-0.4089	-7.01
4.170	0.1780	-0.4083	-7.03
4.175	0.1776	-0.4080	-7.03
4.180	0.1769	-0.4076	-7.04
4.185	0.1764	-0.4075	-7.05
4.190	0.1758	-0.4073	-7.06
4.195	0.1753	-0.4072	-7.06
4.200	0.1749	-0.4071	-7.07
4.205	0.1745	-0.4073	-7.07
4.210	0.1741	-0.4071	-7.08
4.215	0.1737	-0.4071	-7.08
4.220	0.1733	-0.4071	-7.08
4.225	0.1727	-0.4070	-7.09
4.230	0.1724	-0.4071	-7.09
4.235	0.1721	-0.4071	-7.09
4.240	0.1717	-0.4071	-7.10
4.245	0.1715	-0.4073	-7.09
4.250	0.1711	-0.4074	-7.09
4.255	0.1709	-0.4073	-7.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.260	0.1705	-0.4074	-7.10
4.265	0.1703	-0.4075	-7.10
4.270	0.1702	-0.4075	-7.10
4.275	0.1701	-0.4076	-7.10
4.280	0.1699	-0.4076	-7.10
4.285	0.1699	-0.4077	-7.10
4.290	0.1696	-0.4078	-7.10
4.295	0.1695	-0.4078	-7.10
4.300	0.1695	-0.4078	-7.10
4.305	0.1694	-0.4079	-7.10
4.310	0.1692	-0.4079	-7.10
4.315	0.1690	-0.4079	-7.10
4.320	0.1688	-0.4080	-7.10
4.325	0.1687	-0.4082	-7.10
4.330	0.1685	-0.4082	-7.10
4.335	0.1684	-0.4083	-7.10
4.340	0.1682	-0.4087	-7.09
4.345	0.1682	-0.4088	-7.09
4.350	0.1681	-0.4091	-7.09
4.355	0.1680	-0.4095	-7.08
4.360	0.1681	-0.4098	-7.07
4.365	0.1683	-0.4101	-7.07
4.370	0.1683	-0.4104	-7.06
4.375	0.1686	-0.4107	-7.05
4.380	0.1689	-0.4111	-7.04
4.385	0.1689	-0.4113	-7.04
4.390	0.1694	-0.4116	-7.03
4.395	0.1699	-0.4118	-7.02
4.400	0.1701	-0.4121	-7.02
4.405	0.1707	-0.4121	-7.01
4.410	0.1709	-0.4121	-7.01
4.415	0.1714	-0.4121	-7.01
4.420	0.1715	-0.4122	-7.00
4.425	0.1719	-0.4121	-7.00
4.430	0.1723	-0.4120	-7.00
4.435	0.1724	-0.4119	-7.00
4.440	0.1724	-0.4118	-7.00
4.445	0.1726	-0.4117	-7.00
4.450	0.1726	-0.4116	-7.01
4.455	0.1727	-0.4116	-7.01
4.460	0.1730	-0.4114	-7.01
4.465	0.1732	-0.4116	-7.00
4.470	0.1737	-0.4115	-7.00
4.475	0.1742	-0.4119	-6.99
4.480	0.1747	-0.4123	-6.98
4.485	0.1751	-0.4128	-6.97

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.490	0.1757	-0.4132	-6.95
4.495	0.1762	-0.4137	-6.94
4.500	0.1769	-0.4142	-6.93
4.505	0.1774	-0.4146	-6.92
4.510	0.1781	-0.4149	-6.91
4.515	0.1786	-0.4150	-6.90
4.520	0.1794	-0.4151	-6.89
4.525	0.1800	-0.4150	-6.89
4.530	0.1804	-0.4149	-6.89
4.535	0.1806	-0.4148	-6.89
4.540	0.1809	-0.4144	-6.89
4.545	0.1810	-0.4141	-6.90
4.550	0.1812	-0.4140	-6.90
4.555	0.1814	-0.4137	-6.90
4.560	0.1814	-0.4135	-6.91
4.565	0.1816	-0.4132	-6.91
4.570	0.1818	-0.4130	-6.91
4.575	0.1817	-0.4128	-6.92
4.580	0.1818	-0.4124	-6.92
4.585	0.1820	-0.4120	-6.93
4.590	0.1818	-0.4119	-6.93
4.595	0.1817	-0.4116	-6.94
4.600	0.1817	-0.4114	-6.94
4.605	0.1815	-0.4110	-6.95
4.610	0.1812	-0.4106	-6.96
4.615	0.1810	-0.4104	-6.96
4.620	0.1808	-0.4100	-6.97
4.625	0.1808	-0.4096	-6.98
4.630	0.1805	-0.4096	-6.98
4.635	0.1804	-0.4092	-6.99
4.640	0.1803	-0.4089	-7.00
4.645	0.1803	-0.4084	-7.00
4.650	0.1801	-0.4085	-7.00
4.655	0.1797	-0.4080	-7.02
4.660	0.1798	-0.4079	-7.02
4.665	0.1798	-0.4073	-7.03
4.670	0.1799	-0.4073	-7.03
4.675	0.1800	-0.4070	-7.03
4.680	0.1801	-0.4067	-7.04
4.685	0.1803	-0.4066	-7.04
4.690	0.1801	-0.4059	-7.05
4.695	0.1800	-0.4059	-7.05
4.700	0.1803	-0.4055	-7.06
4.705	0.1802	-0.4049	-7.07
4.710	0.1801	-0.4048	-7.07
4.715	0.1800	-0.4041	-7.08

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.720	0.1798	-0.4036	-7.09
4.725	0.1795	-0.4033	-7.10
4.730	0.1794	-0.4029	-7.11
4.735	0.1790	-0.4022	-7.13
4.740	0.1785	-0.4017	-7.14
4.745	0.1776	-0.4011	-7.16
4.750	0.1771	-0.4008	-7.17
4.755	0.1765	-0.4003	-7.18
4.760	0.1758	-0.4000	-7.19
4.765	0.1750	-0.3998	-7.20
4.770	0.1745	-0.3995	-7.21
4.775	0.1740	-0.3993	-7.22
4.780	0.1737	-0.3990	-7.23
4.785	0.1736	-0.3990	-7.23
4.790	0.1732	-0.3986	-7.24
4.795	0.1734	-0.3982	-7.24
4.800	0.1735	-0.3978	-7.25
4.805	0.1736	-0.3976	-7.25
4.810	0.1737	-0.3969	-7.27
4.815	0.1738	-0.3963	-7.28
4.820	0.1741	-0.3956	-7.29
4.825	0.1742	-0.3950	-7.30
4.830	0.1741	-0.3946	-7.30
4.835	0.1742	-0.3941	-7.31
4.840	0.1743	-0.3936	-7.32
4.845	0.1742	-0.3933	-7.33
4.850	0.1741	-0.3929	-7.34
4.855	0.1742	-0.3928	-7.34
4.860	0.1740	-0.3926	-7.34
4.865	0.1737	-0.3924	-7.35
4.870	0.1738	-0.3924	-7.35
4.875	0.1737	-0.3925	-7.35
4.880	0.1736	-0.3924	-7.35
4.885	0.1735	-0.3926	-7.35
4.890	0.1733	-0.3928	-7.34
4.895	0.1732	-0.3929	-7.34
4.900	0.1732	-0.3930	-7.34
4.905	0.1731	-0.3932	-7.34
4.910	0.1731	-0.3932	-7.34
4.915	0.1729	-0.3935	-7.33
4.920	0.1728	-0.3937	-7.33
4.925	0.1727	-0.3938	-7.33
4.930	0.1726	-0.3940	-7.33
4.935	0.1724	-0.3942	-7.32
4.940	0.1723	-0.3944	-7.32
4.945	0.1723	-0.3946	-7.32

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	0.1722	-0.3947	-7.32
4.955	0.1720	-0.3949	-7.32
4.960	0.1718	-0.3951	-7.31
4.965	0.1718	-0.3952	-7.31
4.970	0.1717	-0.3954	-7.31
4.975	0.1715	-0.3953	-7.31
4.980	0.1714	-0.3955	-7.31
4.985	0.1712	-0.3957	-7.31
4.990	0.1711	-0.3956	-7.31
4.995	0.1711	-0.3958	-7.31
5.000	0.1710	-0.3958	-7.31
5.005	0.1711	-0.3958	-7.31
5.010	0.1711	-0.3959	-7.31
5.015	0.1713	-0.3958	-7.30
5.020	0.1714	-0.3958	-7.30
5.025	0.1717	-0.3960	-7.30
5.030	0.1720	-0.3957	-7.30
5.035	0.1724	-0.3959	-7.29
5.040	0.1726	-0.3957	-7.30
5.045	0.1731	-0.3958	-7.29
5.050	0.1736	-0.3957	-7.29
5.055	0.1739	-0.3959	-7.28
5.060	0.1742	-0.3957	-7.28
5.065	0.1746	-0.3960	-7.28
5.070	0.1750	-0.3957	-7.28
5.075	0.1750	-0.3957	-7.28
5.080	0.1754	-0.3955	-7.28
5.085	0.1754	-0.3955	-7.28
5.090	0.1758	-0.3953	-7.28
5.095	0.1756	-0.3951	-7.28
5.100	0.1758	-0.3949	-7.29
5.105	0.1755	-0.3947	-7.29
5.110	0.1756	-0.3946	-7.29
5.115	0.1756	-0.3944	-7.30
5.120	0.1757	-0.3942	-7.30
5.125	0.1755	-0.3941	-7.30
5.130	0.1756	-0.3940	-7.30
5.135	0.1756	-0.3936	-7.31
5.140	0.1756	-0.3934	-7.31
5.145	0.1757	-0.3933	-7.32
5.150	0.1755	-0.3931	-7.32
5.155	0.1754	-0.3927	-7.33
5.160	0.1753	-0.3926	-7.33
5.165	0.1751	-0.3923	-7.34
5.170	0.1750	-0.3920	-7.35
5.175	0.1747	-0.3918	-7.35

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	0.1745	-0.3914	-7.36
5.185	0.1742	-0.3908	-7.37
5.190	0.1738	-0.3905	-7.38
5.195	0.1735	-0.3901	-7.39
5.200	0.1731	-0.3897	-7.40
5.205	0.1727	-0.3892	-7.42
5.210	0.1722	-0.3888	-7.43
5.215	0.1718	-0.3885	-7.44
5.220	0.1714	-0.3879	-7.45
5.225	0.1710	-0.3873	-7.46
5.230	0.1703	-0.3867	-7.48
5.235	0.1699	-0.3861	-7.50
5.240	0.1694	-0.3855	-7.51
5.245	0.1687	-0.3848	-7.53
5.250	0.1682	-0.3843	-7.55
5.255	0.1676	-0.3835	-7.57
5.260	0.1670	-0.3830	-7.58
5.265	0.1662	-0.3823	-7.60
5.270	0.1655	-0.3818	-7.62
5.275	0.1648	-0.3812	-7.63
5.280	0.1640	-0.3805	-7.65
5.285	0.1634	-0.3800	-7.67
5.290	0.1629	-0.3796	-7.68
5.295	0.1619	-0.3793	-7.69
5.300	0.1615	-0.3786	-7.71
5.305	0.1607	-0.3782	-7.72
5.310	0.1600	-0.3776	-7.74
5.315	0.1594	-0.3772	-7.75
5.320	0.1590	-0.3767	-7.77
5.325	0.1583	-0.3763	-7.78
5.330	0.1576	-0.3759	-7.79
5.335	0.1572	-0.3753	-7.81
5.340	0.1566	-0.3749	-7.82
5.345	0.1561	-0.3743	-7.84
5.350	0.1555	-0.3738	-7.85
5.355	0.1549	-0.3731	-7.87
5.360	0.1544	-0.3726	-7.89
5.365	0.1536	-0.3720	-7.91
5.370	0.1531	-0.3713	-7.92
5.375	0.1523	-0.3705	-7.95
5.380	0.1515	-0.3697	-7.97
5.385	0.1505	-0.3692	-7.99
5.390	0.1496	-0.3686	-8.01
5.395	0.1486	-0.3682	-8.02
5.400	0.1479	-0.3677	-8.04
5.405	0.1468	-0.3676	-8.05

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.410	0.1460	-0.3671	-8.07
5.415	0.1453	-0.3667	-8.08
5.420	0.1445	-0.3664	-8.09
5.425	0.1439	-0.3661	-8.10
5.430	0.1429	-0.3657	-8.12
5.435	0.1422	-0.3656	-8.13
5.440	0.1415	-0.3653	-8.14
5.445	0.1407	-0.3649	-8.15
5.450	0.1401	-0.3648	-8.16
5.455	0.1396	-0.3646	-8.17
5.460	0.1389	-0.3643	-8.18
5.465	0.1384	-0.3640	-8.19
5.470	0.1377	-0.3636	-8.20
5.475	0.1370	-0.3632	-8.22
5.480	0.1364	-0.3629	-8.23
5.485	0.1357	-0.3626	-8.24
5.490	0.1349	-0.3622	-8.26
5.495	0.1342	-0.3619	-8.27
5.500	0.1333	-0.3616	-8.28
5.505	0.1326	-0.3613	-8.29
5.510	0.1316	-0.3612	-8.30
5.515	0.1309	-0.3611	-8.31
5.520	0.1302	-0.3611	-8.32
5.525	0.1297	-0.3612	-8.32
5.530	0.1288	-0.3613	-8.32
5.535	0.1284	-0.3614	-8.32
5.540	0.1278	-0.3616	-8.32
5.545	0.1273	-0.3616	-8.33
5.550	0.1268	-0.3618	-8.33
5.555	0.1263	-0.3618	-8.33
5.560	0.1259	-0.3619	-8.33
5.565	0.1253	-0.3620	-8.33
5.570	0.1249	-0.3620	-8.34
5.575	0.1243	-0.3621	-8.34
5.580	0.1238	-0.3619	-8.35
5.585	0.1231	-0.3622	-8.35
5.590	0.1226	-0.3622	-8.35
5.595	0.1219	-0.3623	-8.35
5.600	0.1212	-0.3625	-8.35
5.605	0.1205	-0.3629	-8.35
5.610	0.1198	-0.3632	-8.35
5.615	0.1191	-0.3636	-8.34
5.620	0.1186	-0.3640	-8.34
5.625	0.1182	-0.3648	-8.32
5.630	0.1178	-0.3652	-8.32
5.635	0.1174	-0.3656	-8.31

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.640	0.1171	-0.3660	-8.31
5.645	0.1168	-0.3666	-8.30
5.650	0.1165	-0.3669	-8.29
5.655	0.1163	-0.3672	-8.29
5.660	0.1160	-0.3676	-8.28
5.665	0.1157	-0.3678	-8.28
5.670	0.1154	-0.3681	-8.27
5.675	0.1152	-0.3687	-8.26
5.680	0.1147	-0.3692	-8.26
5.685	0.1144	-0.3695	-8.25
5.690	0.1142	-0.3698	-8.25
5.695	0.1141	-0.3701	-8.24
5.700	0.1140	-0.3704	-8.23
5.705	0.1138	-0.3707	-8.23
5.710	0.1136	-0.3711	-8.22
5.715	0.1136	-0.3713	-8.22
5.720	0.1134	-0.3715	-8.21
5.725	0.1133	-0.3719	-8.21
5.730	0.1132	-0.3722	-8.20
5.735	0.1131	-0.3723	-8.20
5.740	0.1130	-0.3725	-8.19
5.745	0.1130	-0.3726	-8.19
5.750	0.1129	-0.3729	-8.19
5.755	0.1129	-0.3729	-8.19
5.760	0.1129	-0.3730	-8.18
5.765	0.1128	-0.3732	-8.18
5.770	0.1128	-0.3731	-8.18
5.775	0.1127	-0.3731	-8.19
5.780	0.1127	-0.3729	-8.19
5.785	0.1126	-0.3730	-8.19
5.790	0.1127	-0.3727	-8.19
5.795	0.1123	-0.3725	-8.20
5.800	0.1123	-0.3723	-8.20
5.805	0.1122	-0.3723	-8.21
5.810	0.1121	-0.3719	-8.21
5.815	0.1119	-0.3717	-8.22
5.820	0.1118	-0.3713	-8.23
5.825	0.1115	-0.3709	-8.24
5.830	0.1114	-0.3707	-8.24
5.835	0.1111	-0.3702	-8.26
5.840	0.1110	-0.3698	-8.27
5.845	0.1105	-0.3694	-8.28
5.850	0.1106	-0.3690	-8.29
5.855	0.1100	-0.3686	-8.30
5.860	0.1098	-0.3680	-8.31
5.865	0.1094	-0.3675	-8.33

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	0.1090	-0.3671	-8.34
5.875	0.1085	-0.3668	-8.35
5.880	0.1080	-0.3664	-8.36
5.885	0.1076	-0.3661	-8.37
5.890	0.1070	-0.3657	-8.38
5.895	0.1069	-0.3653	-8.39
5.900	0.1065	-0.3650	-8.40
5.905	0.1058	-0.3647	-8.41
5.910	0.1056	-0.3643	-8.42
5.915	0.1051	-0.3640	-8.43
5.920	0.1047	-0.3636	-8.44
5.925	0.1044	-0.3632	-8.45
5.930	0.1040	-0.3627	-8.47
5.935	0.1036	-0.3620	-8.48
5.940	0.1034	-0.3616	-8.49
5.945	0.1029	-0.3611	-8.51
5.950	0.1025	-0.3607	-8.52
5.955	0.1020	-0.3600	-8.54
5.960	0.1015	-0.3595	-8.55
5.965	0.1007	-0.3587	-8.58
5.970	0.1000	-0.3584	-8.59
5.975	0.0992	-0.3580	-8.60
5.980	0.0983	-0.3578	-8.61
5.985	0.0974	-0.3575	-8.62
5.990	0.0968	-0.3573	-8.63
5.995	0.0960	-0.3574	-8.63
6.000	0.0954	-0.3574	-8.64

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.